ECG RHYTHM REFERENCE

Recognise it, then act on it. 68 rhythms, each on a strip with criteria, features to look for and management.

SmartCare
AHA 2020 + 2025 update
2026-10



READ Every strip in seven steps

1. **Patient first**
   Pulse, perfusion, symptoms. Treat the patient, not the monitor; an organised rhythm can still be PEA.
2. **Rate**
   Regular: 300 ÷ large boxes (or 1500 ÷ small boxes) between R waves. Irregular: QRS count in 6 s × 10.
3. **Rhythm**
   Regular, regularly irregular (grouped beats, bigeminy) or irregularly irregular.
4. **P waves**
   Present? Upright in II? One before every QRS, and a QRS after every P?
5. **PR interval**
   120–200 ms. Constant, lengthening, short, or no relationship to the QRS?
6. **QRS width**
   Narrow (< 120 ms): supraventricular. Wide (≥ 120 ms): ventricular, bundle-branch block, paced, pre-excited or hyperkalaemia.
7. **ST, T and QT**
   ST elevation or depression and its territory; T-wave shape; QTc (> 500 ms carries a high risk of torsades).

SORT Fast tachycardia triage

	Regular	Irregular
Narrow	Sinus tachycardia · SVT (AVNRT/AVRT) · flutter 2:1 · atrial tachycardia	AF · flutter with variable block · MAT · frequent PACs
Wide	**VT until proven otherwise** · SVT with aberrancy · paced · hyperkalaemia	AF with BBB · **pre-excited AF (no AV-nodal blockers)** · polymorphic VT / torsades

NORM Normal values

Paper	25 mm/s · 10 mm/mV	1 small box 40 ms / 0.1 mV; 1 large box 200 ms / 0.5 mV
Heart rate	60–100 /min	< 60 bradycardia · > 100 tachycardia
P wave	< 120 ms	< 2.5 mm tall in II; upright in I, II, aVF
PR interval	120–200 ms	Short: pre-excitation or junctional. Long: AV block
QRS	< 120 ms	110–119 ms: incomplete block
QTc	≤ 450 / 460 ms	Men / women. > 500 ms: high risk. Fridericia QT ÷ ∛RR
Axis	−30° to +90°	LAD < −30° · RAD > +90°
ST segment	Isoelectric	STEMI: ≥ 1 mm in 2 contiguous leads (V2–V3: ≥ 2 mm men ≥ 40 y, ≥ 2.5 mm men < 40 y, ≥ 1.5 mm women)

KEY How each card reads

- **Stable** No immediate threat
- **Assess** Assess the patient and the cause
- **Urgent** Potentially unstable — act promptly
- **Arrest** No cardiac output — resuscitate

Do now · First line · Escalate · ✕ Avoid

Rhythm strips: 7 s at true scale (25 mm/s, 10 mm/mV) with a 1 mV calibration pulse. ACS, conduction and electrolyte patterns are shown as 12-leads at 74% scale; the lead to study is named on the card.

FIND Contents

SINUS	NSR · Sinus brady · Sinus tach · Sinus arrhythmia · Sinus pause
ATRIAL & SVT	PACs · MAT · AF · AF with RVR · Flutter · Flutter 2:1 · SVT · PSVT · Pre-excited AF
JUNCTIONAL	Junctional · Accel. junctional
VENTRICULAR	PVCs · Bigeminy · Multiform PVCs · NSVT · IVR · AIVR · VT · Polymorphic VT · Torsades
CARDIAC ARREST	Coarse VF · Fine VF · Pulseless VT · PEA · Asystole · P-wave asystole · Agonal
AV BLOCK	1° AV block · Mobitz I · Mobitz II · 2:1 block · High-grade block · CHB
CONDUCTION	RBBB · LBBB · LAFB · WPW
PACEMAKER	V-paced · AV-paced · Failure to capture · Undersensing
ACS & ISCHAEMIA	**MI location map** · Inferior STEMI · Inferior + RV STEMI · Anterior STEMI · Septal STEMI · Lateral STEMI · Proximal LAD STEMI · Posterior STEMI · Diffuse ST depression · Wellens · de Winter · Pericarditis
ELECTROLYTES & OTHER	Hyperkalaemia · Hypokalaemia · Hypocalcaemia · Hypercalcaemia · Digoxin effect · Long QT · Brugada · LVH · PE pattern · Hypothermia · Electrical alternans

A study aid, not a substitute for clinical judgement. Management follows the AHA Guidelines for CPR and ECC (2020) and the 2025 update. Local protocols and medical direction always take precedence. Strips are generated by the SmartCare ECG engine.

ALGORITHMS AT A GLANCE

Cardiac arrest — shockable (VF / pulseless VT)

1. **Start CPR**
   Push hard (5–6 cm) and fast (100–120/min) with full recoil; minimise pauses. Give oxygen and attach the monitor/defibrillator.
2. **Shock**
   Biphasic at the manufacturer's recommended energy (typically 120–200 J; if unknown, use the maximum). Resume CPR immediately for 2 minutes.
3. **IV / IO access**
   IV first; IO if IV cannot be obtained quickly. The endotracheal drug route is no longer recommended.
4. **Shock → CPR → epinephrine**
   Epinephrine 1 mg IV/IO every 3–5 minutes. Consider an advanced airway and waveform capnography.
5. **Shock → CPR → antiarrhythmic**
   Amiodarone 300 mg, then 150 mg — or lidocaine 1–1.5 mg/kg, then 0.5–0.75 mg/kg.
6. **Treat reversible causes**
   Hypovolaemia, hypoxia, H^+ (acidosis), hypo-/hyperkalaemia, hypothermia; tension pneumothorax, tamponade, toxins, thrombosis (pulmonary or coronary).
7. **Refractory VF (≥ 3 shocks)**
   A defibrillation vector change (anterior–posterior pads) or double sequential defibrillation may be considered.

Cardiac arrest — non-shockable (PEA / asystole)

1. **Start CPR**
   High-quality CPR with minimal interruptions; oxygen; attach the monitor.
2. **Epinephrine as soon as possible**
   1 mg IV/IO, then every 3–5 minutes.
3. **CPR 2 min → rhythm check**
   Shockable? Switch to the VF/pVT pathway. Organised rhythm? Check for a pulse (≤ 10 s).
4. **Treat reversible causes**
   Hypovolaemia, hypoxia, H^+ (acidosis), hypo-/hyperkalaemia, hypothermia; tension pneumothorax, tamponade, toxins, thrombosis (pulmonary or coronary). Point-of-care ultrasound must not prolong pauses.
5. **Airway and capnography**
   With an advanced airway give 1 breath every 6 s with continuous compressions; $ETCO_2$ tracks CPR quality and can signal ROSC.

Bradycardia with a pulse

1. **Assess and support**
   Airway, oxygen if hypoxaemic, monitor, IV, 12-lead. Is the slow rate causing hypotension, altered mental status, shock, ischaemic chest pain or acute heart failure?
2. **Atropine 1 mg IV**
   Repeat every 3–5 min to a maximum of 3 mg. Often ineffective for infranodal (Mobitz II / complete) block and after heart transplant.
3. **If atropine is ineffective**
   Transcutaneous pacing and/or epinephrine 2–10 mcg/min or dopamine 5–20 mcg/kg/min, titrated to response.
4. **Expert consultation**
   Transvenous pacing; treat the cause (drugs, ischaemia, hyperkalaemia).

Tachycardia with a pulse

1. **Assess and support**
   A rate ≥ 150/min is typical when the arrhythmia itself is the problem. Oxygen if hypoxaemic, monitor, IV, 12-lead ECG.
2. **Unstable → synchronized cardioversion**
   Hypotension, altered mental status, shock, ischaemic chest pain or acute heart failure. Sedate if possible. Common first energies: narrow regular or monomorphic VT ~100 J; AF and flutter ≥ 200 J biphasic.
3. **Stable, narrow and regular**
   Vagal manoeuvres → adenosine 6 mg rapid IV push with flush, then 12 mg → β-blocker or calcium-channel blocker.
4. **Stable, wide QRS (≥ 120 ms)**
   Adenosine only if regular and monomorphic. Antiarrhythmic infusion (procainamide or amiodarone) and expert consultation.
5. **Irregular narrow (AF)**
   Rate control with a β-blocker or diltiazem; assess stroke risk and anticoagulation.

Acute coronary syndrome

1. **12-lead ECG within 10 minutes**
   Notify the receiving centre early; repeat ECGs if symptoms persist.
2. **Aspirin 162–325 mg chewed**
   Unless true allergy or active bleeding.
3. **STEMI → reperfusion**
   Primary PCI with first medical contact-to-device ≤ 90 min (≤ 120 min if transfer is needed); fibrinolysis within 30 min of arrival if PCI cannot be achieved in time.
4. **Symptom relief**
   Nitroglycerin if SBP ≥ 90 mmHg with no RV infarction or recent PDE-5 inhibitor; oxygen only if SpO_2 < 90%; opioid for refractory pain.

SINUS

5 rhythms · 25 mm/s, 10 mm/mV

Normal sinus rhythm STABLE

The normal heartbeat: a sinus P wave before every QRS at 60–100/min.



CRITERIA

Rate	60–100/min
Rhythm	Regular (slight respiratory variation is normal)
P waves	Upright in I, II, aVF — one before every QRS
PR	120–200 ms, constant
QRS	< 120 ms

LOOK FOR

- Upright P in II before every QRS, and a QRS after every P
- Constant PR interval
- R–R intervals vary by less than about 10%

ACT

- **No rhythm treatment needed** — Treat the patient, not the monitor: correlate with pulse, perfusion and symptoms

Don't confuse with ≠ **Sinus arrhythmia** identical P waves, but R–R varies with breathing ≠ **1st-degree block** PR longer than 200 ms

Sinus bradycardia ASSESS

Sinus rhythm slower than 60/min.



CRITERIA

Rate	< 60/min
Rhythm	Regular
P waves	Normal, one per QRS
PR	120–200 ms (may be slightly long)
QRS	< 120 ms

LOOK FOR

- Normal P–QRS–T complexes spaced far apart
- Long flat baseline between the T wave and the next P
- P waves identical to sinus rhythm

ACT

- **Look for the cause** — Hypoxia (especially in children), drugs, hypothermia, inferior MI, raised ICP
- **Atropine 1 mg IV if symptomatic** — Repeat every 3–5 min to a maximum of 3 mg
- **Pacing or chronotropic infusion** — If atropine fails: transcutaneous pacing, epinephrine 2–10 mcg/min or dopamine 5–20 mcg/kg/min

Don't confuse with ≠ **Junctional rhythm** no upright P before the QRS ≠ **2:1 AV block** a second, non-conducted P hides near each T wave

Sinus tachycardia ASSESS

Sinus rhythm faster than 100/min — almost always a response to something else.



CRITERIA

Rate	> 100/min (rarely > 220 − age)
Rhythm	Regular; rate changes gradually
P waves	Normal; may run into the preceding T wave
PR	Normal or slightly short
QRS	< 120 ms

LOOK FOR

- Upright P before each QRS — look at the end of the T wave
- Rate speeds and slows gradually, never abruptly
- Usually < 150/min at rest in adults

ACT

- **Find and treat the cause** — Oxygenation, volume, temperature, pain, sepsis, PE, drugs

× Do not cardiovert or give adenosine

Don't confuse with ≠ **SVT** abrupt onset, fixed rate usually > 150/min, no visible P waves ≠ **Flutter 2:1** rate fixed near 150/min with flutter waves in II, III, aVF

Sinus arrhythmia STABLE

Sinus rhythm whose rate rises with inspiration and falls with expiration.



CRITERIA

Rate	Usually 60–100/min, varying
Rhythm	Irregular in a smooth, cyclical pattern
P waves	Identical, one per QRS
PR	Constant
QRS	< 120 ms

LOOK FOR

- P waves all the same shape — only the spacing changes
- R–R shortens and lengthens smoothly every few beats, in time with breathing
- Constant PR interval

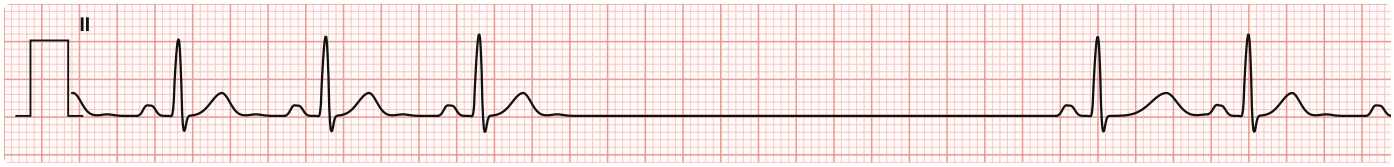
ACT

- **No treatment** — Confirm that the variation follows the breathing pattern

Don't confuse with ≠ **AF** no P waves and no repeating pattern ≠ **MAT** at least three different P-wave shapes

Sinus pause / arrest ASSESS

The sinus node stops firing for a moment — a sudden gap with no P wave.



CRITERIA

Rate	Underlying rate normal
Rhythm	Irregular — abrupt pauses
P waves	Absent during the pause
Pause	Not an exact multiple of the P–P interval
QRS	< 120 ms

LOOK FOR

- A sudden flat gap with no P wave
- Pause length unrelated to the P–P interval (an exact multiple suggests SA exit block)
- Measure the pause with calipers: > 3 s is significant

ACT

- **Correlate with symptoms** — Dizziness or syncope during pauses?
- **Symptomatic → bradycardia algorithm** — Atropine, then pacing if unstable

Don't confuse with ≠ **2:1 block** a P wave is present but not conducted ≠ **Blocked PAC** an early P (often in the T wave) precedes the pause

ATRIAL & SVT

9 rhythms · 25 mm/s, 10 mm/mV

Premature atrial complexes STABLE

Early beats from an atrial focus, each with a differently shaped P wave.



CRITERIA

Rate	Underlying rate
Rhythm	Irregular at each early beat
P waves	Premature and abnormal in shape (may hide in the T)
PR	May differ from sinus
QRS	Usually narrow, like the sinus QRS

LOOK FOR

- An early QRS preceded by an abnormal P wave
- The pause after it is less than fully compensatory (the sinus node is reset)
- The QRS looks like the sinus QRS

ACT

- **Usually no treatment** — Reduce triggers

Don't confuse with ≠ **PVC** wide QRS, no preceding P, fully compensatory pause ≠ **Sinus arrhythmia** gradual change, identical P waves

Multifocal atrial tachycardia ASSESS

A fast, irregular rhythm with at least three different P-wave shapes.



CRITERIA

Rate	> 100/min
Rhythm	Irregularly irregular
P waves	≥ 3 different morphologies
PR	Varies with the P-wave focus
QRS	< 120 ms

LOOK FOR

- Every QRS has a P in front of it — but the P waves keep changing shape
- Variable PR and R–R intervals
- An isoelectric baseline between P waves (unlike AF)

ACT

- **Treat the underlying illness** — Oxygenation, bronchodilation, sepsis
- ✕ **Do not cardiovert**
- **Rate control if needed** — Calcium-channel blocker under expert guidance

Don't confuse with ≠ **AF** no discrete P waves — chaotic baseline ≠ **Sinus tachycardia** every P wave identical

Atrial fibrillation ASSESS

Chaotic atrial activity with an irregularly irregular ventricular response.



CRITERIA

Atrial	350–600/min fibrillatory (f) waves
Ventricular	60–110/min when controlled
Rhythm	Irregularly irregular
P waves	Absent — undulating baseline
QRS	Narrow unless aberrancy or BBB

LOOK FOR

- No P waves — a wavy, uneven baseline, often best in V1
- R–R intervals with no pattern at all
- Pulse deficit: some beats come too early to produce a pulse (compare HR with pulse)

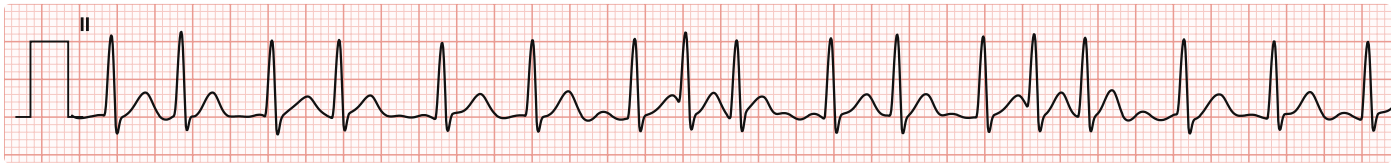
ACT

- **Assess stability and onset** — Onset < 48 h or ≥ 48 h/unknown determines whether cardioversion is safe without anticoagulation
- **Rate control if the rate rises** — β-blocker or diltiazem if not in decompensated heart failure

Don't confuse with ≠ **MAT** distinct, changing P waves before each QRS ≠ **Flutter with variable block** organised sawtooth waves

AF with rapid ventricular response URGENT

Atrial fibrillation with a ventricular rate above 110/min.



CRITERIA

Ventricular	> 110/min (often 120–180)
Rhythm	Irregularly irregular
P waves	Absent
QRS	Usually narrow

LOOK FOR

- Fast and irregularly irregular — measure several R–R intervals
- At high rates the irregularity can look almost regular
- No consistent P waves

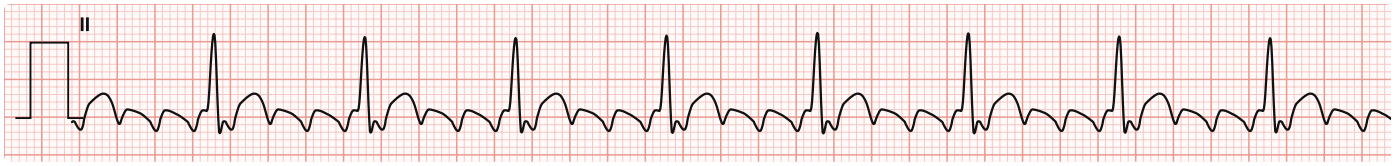
ACT

- **Unstable → synchronized cardioversion** — Initial biphasic energy of at least 200 J is reasonable (2025 AHA)
- **Stable → rate control** — Diltiazem 0.25 mg/kg IV over 2 min (≈ 15–20 mg) or a β-blocker such as metoprolol 2.5–5 mg IV
- **Amiodarone 150 mg IV over 10 min** — Option in heart failure or when other agents are contraindicated

Don't confuse with ≠ **Pre-excited AF** wide, bizarre, varying QRS often > 200/min — no AV-nodal blockers ≠ **MAT** different P waves before each QRS

Atrial flutter (4:1) ASSESS

A regular atrial sawtooth at about 300/min with a fixed AV conduction ratio.



CRITERIA

Atrial	250–350/min (typically 300)
Ventricular	Atrial rate ÷ ratio (4:1 ≈ 75/min)
Rhythm	Regular (irregular if the ratio varies)
P waves	Flutter (F) waves: sawtooth in II, III, aVF; upright in V1
QRS	Narrow

LOOK FOR

- Continuous sawtooth with no flat baseline in II, III and aVF
- Count F waves between QRS complexes to find the ratio
- Some F waves hide inside the QRS and T waves

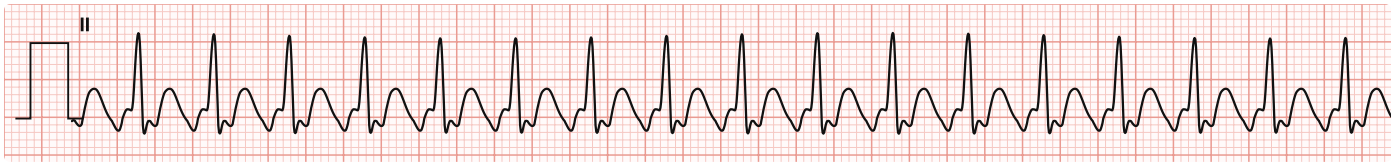
ACT

- **Unstable → synchronized cardioversion** — Initial biphasic energy of at least 200 J (2025 AHA)
- **Rate control** — β-blocker or diltiazem — flutter is often harder to rate-control than AF

Don't confuse with ≠ **AF** irregularly irregular with a chaotic baseline ≠ **2:1 flutter** ~150/min, F waves easily missed

Atrial flutter with 2:1 block URGENT

Flutter conducting every second wave — a regular narrow tachycardia locked near 150/min.



CRITERIA

Atrial	~300/min
Ventricular	~150/min
Rhythm	Regular
F waves	One visible between QRS complexes, the other hidden in the QRS/T
QRS	Narrow

LOOK FOR

- A regular narrow tachycardia at ~150/min — always consider flutter
- Look for a second F wave at the end of the QRS or in the T
- Vagal manoeuvres or adenosine increase the block and unmask the sawtooth

ACT

- **Unstable → synchronized cardioversion** — Initial biphasic energy of at least 200 J (2025 AHA)
- **Diagnostic adenosine** — 6 mg rapid IV push, recording a rhythm strip, to reveal F waves
- **Rate control and stroke-risk assessment** — β-blocker or diltiazem

Don't confuse with ≠ **SVT** often 180–220/min; adenosine terminates it ≠ **Sinus tachycardia** upright P, rate varies

Supraventricular tachycardia (AVNRT) URGENT

A regular narrow-complex tachycardia — usually re-entry within the AV node.



CRITERIA

Rate	150–250/min
Rhythm	Perfectly regular
P waves	Not visible, or retrograde just after the QRS (pseudo-r′ in V1)
PR	Not measurable
QRS	Narrow

LOOK FOR

- A fixed rate that does not vary
- No P waves before the QRS — look for a small notch at the end of the QRS
- Abrupt onset and offset

ACT

- **Unstable → synchronized cardioversion** — Narrow regular: start around 100 J biphasic (device-specific), escalate
- **Vagal manoeuvres** — Modified Valsalva: strain for 15 s, then lie flat with the legs raised
- **Adenosine** — 6 mg rapid IV push with a 20 mL flush
- **Then a calcium-channel blocker or β-blocker** — Diltiazem or verapamil, or a β-blocker, if adenosine fails or SVT recurs

Don't confuse with ≠ **Sinus tachycardia** visible P waves, gradual rate change ≠ **Flutter 2:1** ~150/min with hidden F waves

Paroxysmal SVT URGENT

SVT that switches on and off suddenly — here triggered by a premature atrial beat.



CRITERIA

Rate	Sinus, then 150–250/min
Onset	Abrupt, after a PAC
Termination	Abrupt, often with a brief pause
QRS	Narrow

LOOK FOR

- A premature P wave with a longer PR starts the run
- Instant jump to a fixed fast rate
- Sudden termination, sometimes with a pause before sinus rhythm resumes

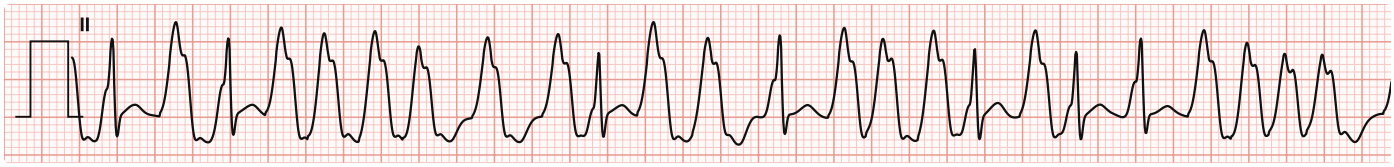
ACT

- **Unstable → synchronized cardioversion** — Around 100 J biphasic to start
- **Vagal manoeuvres, then adenosine** — Adenosine 6 mg rapid IV push, then 12 mg
- **Recurrent episodes** — Calcium-channel blocker or β-blocker

Don't confuse with ≠ **SVT** the sustained form ≠ **PACs** single early beats without a run

Pre-excited AF (AF with WPW) URGENT

AF conducting down an accessory pathway: fast, broad and irregular — and dangerous.



CRITERIA

Rate	Often > 200/min
Rhythm	Irregularly irregular
P waves	Absent
QRS	Wide and varying beat to beat
Shortest R–R	May be < 250 ms

LOOK FOR

- "FBI": fast, broad, irregular
- QRS width and shape change from beat to beat
- Very short R–R intervals in places

ACT

- **Unstable → synchronized cardioversion** — Immediately
- **Stable → procainamide** — 20–50 mg/min IV until the arrhythmia is suppressed, hypotension occurs, the QRS widens > 50% …
- **× Avoid AV-nodal blockers**

Don't confuse with ≠ **AF with RVR** narrow, uniform QRS ≠ **Polymorphic VT** continuously changing axis, usually pulseless

JUNCTIONAL

2 rhythms · 25 mm/s, 10 mm/mV

Junctional escape rhythm ASSESS

The AV junction takes over as pacemaker at 40–60/min.



CRITERIA

Rate	40–60/min
Rhythm	Regular
P waves	Inverted in II/III/aVF just before (PR < 120 ms), hidden in, or after the QRS
QRS	Narrow

LOOK FOR

- Narrow QRS with no normal upright P before it
- Inverted (retrograde) P with a very short PR — or no P at all
- Regular, slow rate

ACT

- **Treat if symptomatic** — Bradycardia algorithm: atropine, then pacing or chronotropic infusion
- **× Do not suppress it**

Don't confuse with ≠ **Sinus bradycardia** upright P with a normal PR ≠ **Idioventricular rhythm** wide QRS, slower

Accelerated junctional rhythm ASSESS

An irritable AV junction firing at 61–100/min, faster than its escape rate.



CRITERIA

Rate	61–100/min
Rhythm	Regular
P waves	Retrograde (inverted) before or after the QRS, or absent
QRS	Narrow

LOOK FOR

- A narrow regular rhythm without upright P waves
- Retrograde P wave in the ST segment
- Rate faster than a junctional escape

ACT

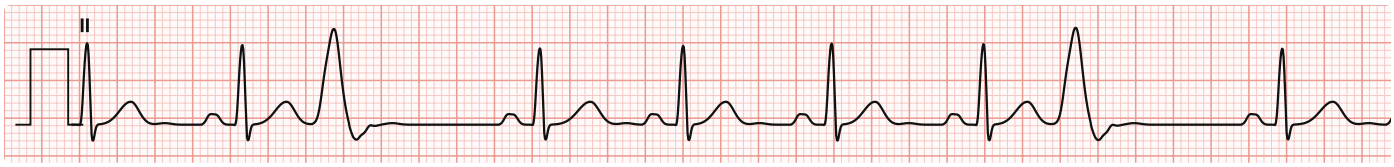
- **Treat the cause** — Check digoxin level and potassium

Don't confuse with ≠ **NSR** upright P before every QRS ≠ **Junctional escape** 40–60/min

VENTRICULAR 9 rhythms · 25 mm/s, 10 mm/mV

Premature ventricular complexes ASSESS

Early, wide, bizarre beats from a ventricular focus.



CRITERIA

Rate	Underlying rate
Rhythm	Irregular at each PVC
P waves	None before the PVC (a sinus P may hide in it)
QRS	≥ 120 ms, bizarre, T wave opposite to the QRS
Pause	Fully compensatory

LOOK FOR

- An early wide QRS with no preceding P wave
- T wave deflects opposite to the main QRS
- Fully compensatory pause: the R–R spanning the PVC equals two normal cycles

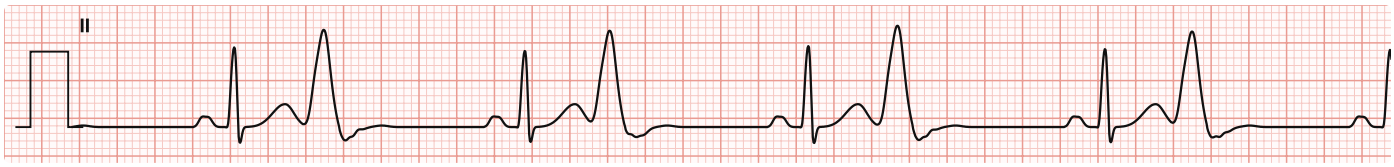
ACT

- **Assess the patient** — Ischaemia? Electrolytes (K^+, Mg^{2+})? Hypoxia? Drugs or stimulants?

Don't confuse with ≠ **PAC** narrow QRS preceded by an early P ≠ **Bigeminy** every second beat is a PVC

Ventricular bigeminy ASSESS

Every second beat is a PVC — normal, PVC, normal, PVC.



CRITERIA

Rate	Underlying rate
Rhythm	Regularly irregular, in pairs
QRS	Alternates narrow (sinus) and wide (PVC)
Pause	After each PVC

LOOK FOR

- A repeating pair of a normal beat followed by a PVC
- Fixed coupling interval to the PVC
- The pulse may be half the monitor rate if the PVCs do not perfuse

ACT

- **Check the pulse against the monitor** — The perfused rate may be half the ECG rate

Don't confuse with ≠ **PVCs** Occasional PVCs ≠ **Multiform PVCs** more than one PVC shape

Multiform PVCs ASSESS

PVCs with two or more different shapes — from more than one focus.



CRITERIA

Rate	Underlying rate
Rhythm	Irregular
QRS	Wide ectopics with different morphologies

LOOK FOR

- PVCs of different shapes in the same lead
- Each ectopic is wide with no preceding P
- Compensatory pauses after each

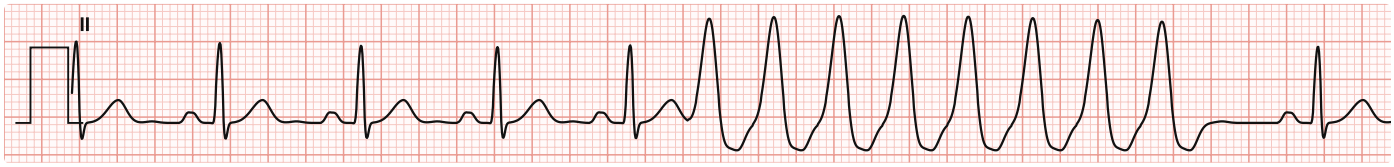
ACT

- **Look for ischaemia and electrolyte disturbance** — Correct K^+ and Mg^{2+}

Don't confuse with ≠ **Unifocal PVCs** all one shape ≠ **Aberrant PACs** preceded by an early P

Non-sustained VT URGENT

Runs of three or more ventricular beats at > 100/min that stop on their own within 30 s.



CRITERIA

Run rate	> 100/min (here ~175)
Duration	≥ 3 beats, < 30 s
QRS	Wide, identical within a run
Between runs	Sinus rhythm

LOOK FOR

- Bursts of identical wide complexes
- Sinus rhythm resumes after each run
- Count the beats and time the run

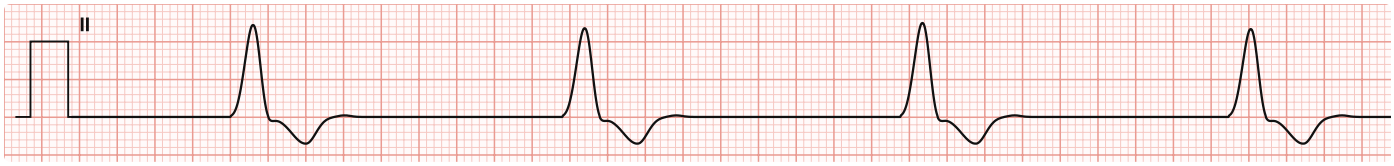
ACT

- **Assess and monitor closely** — Pads on if runs are frequent or long
- **Correct K^+, Mg^{2+} and ischaemia** — Common, reversible triggers
- **Expert consultation** — β-blockers and further work-up (echo, ischaemia)

Don't confuse with ≠ **Sustained VT** > 30 s or causing instability ≠ **PVCs** Isolated PVCs

Idioventricular rhythm URGENT

A ventricular escape rhythm at 20–40/min: wide, slow, no P waves.



CRITERIA

Rate	20–40/min
Rhythm	Regular
P waves	Absent or dissociated
QRS	≥ 120 ms; T opposite to QRS

LOOK FOR

- Slow, wide, regular complexes
- No P waves conducting
- Discordant T waves

ACT

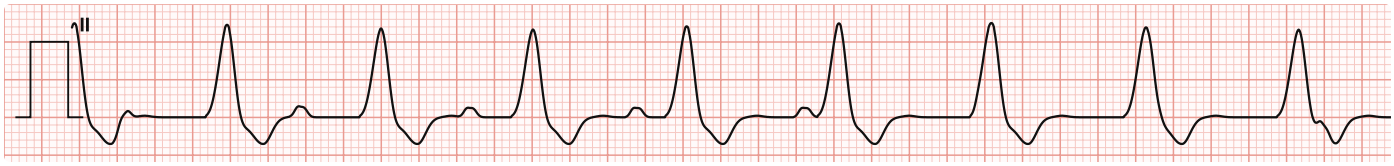
- **Check for a pulse** — Without a pulse this is PEA — start CPR
- **With a pulse → bradycardia algorithm** — Transcutaneous pacing, epinephrine or dopamine infusion

× Never give antiarrhythmics

Don't confuse with ≠ **AIVR** 50–110/min ≠ **CHB** P waves march independently

Accelerated idioventricular rhythm ASSESS

A wide ventricular rhythm at 50–110/min — often a sign of reperfusion.



CRITERIA

Rate	50–110/min
Rhythm	Regular
P waves	Dissociated sinus P waves may march through
QRS	Wide
Onset	Gradual; fusion beats common

LOOK FOR

- A wide regular rhythm at a near-normal rate
- Dissociated P waves; fusion beats as sinus and ventricular rhythms compete
- Usually self-limiting

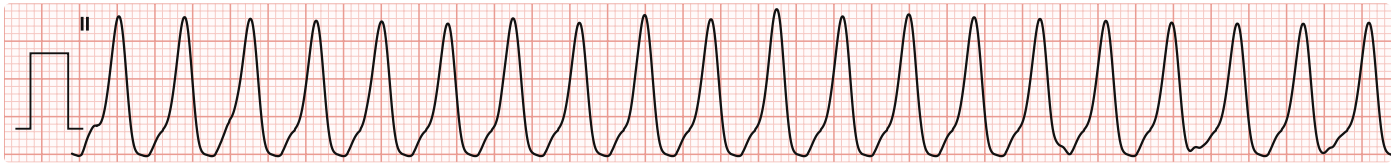
ACT

× Do not treat as VT

Don't confuse with ≠ **VT** usually > 120/min ≠ **IVR** 20–40/min

Monomorphic VT (with pulse) URGENT

A regular wide-complex tachycardia from a single ventricular focus.



CRITERIA

Rate	120–250/min
Rhythm	Regular
P waves	Dissociated — occasionally visible
QRS	≥ 120 ms, identical complexes
Clues	AV dissociation, capture or fusion beats, extreme axis, precordial concordance

LOOK FOR

- Regular, wide, identical complexes
- P waves marching through independently (AV dissociation) — diagnostic
- Extreme axis, precordial concordance or QRS > 160 ms favour VT

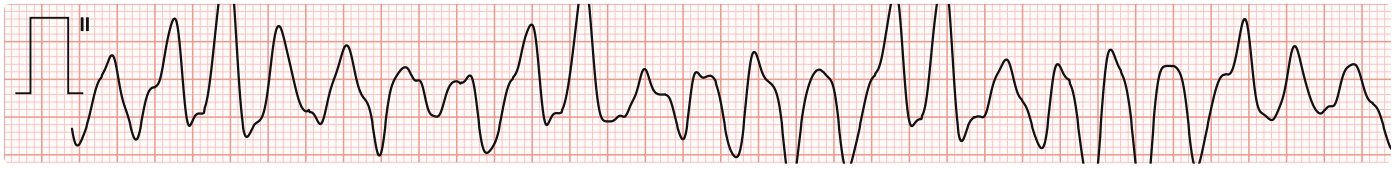
ACT

- **Pulseless? → shockable arrest algorithm** — Defibrillate immediately (unsynchronized)
- **Unstable with a pulse → synchronized cardioversion** — Monomorphic: start around 100 J biphasic and escalate
- **Stable → antiarrhythmic and expert consultation** — Procainamide 20–50 mg/min (max 17 mg/kg

Don't confuse with ≠ **SVT with aberrancy** same rhythm with a BBB — treat as VT if in doubt ≠ **Pre-excited AF** irregular with varying QRS

Polymorphic VT ARREST

A wide-complex tachycardia whose QRS shape changes from beat to beat, with a normal baseline QT.



CRITERIA

Rate	150–300/min
Rhythm	Irregular
QRS	Wide and continuously varying
Baseline QT	Normal (distinguishes it from torsades)

LOOK FOR

- Constantly changing QRS morphology and axis
- Irregular and very fast
- Check the QT before the run: normal QT → polymorphic VT; long QT → torsades

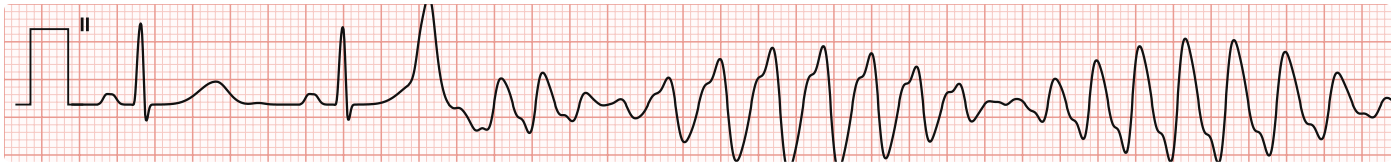
ACT

- **Defibrillate (unsynchronized)** — Synchronization cannot track a varying QRS — use defibrillation energies
- **Treat ischaemia** — The most common cause
- **Antiarrhythmics per arrest algorithm** — Amiodarone or lidocaine

Don't confuse with ≠ **Torsades** long QT and a twisting spindle pattern ≠ **Coarse VF** no identifiable QRS complexes

Torsades de pointes URGENT

Polymorphic VT on a long QT: QRS amplitude waxes and wanes as the axis twists around the baseline.



CRITERIA

Rate	200–250/min
Rhythm	Irregular
QRS	Wide, in spindles that change polarity
Baseline QTc	Prolonged (often > 500 ms)
Onset	Often after a short–long–short sequence (R-on-T PVC)

LOOK FOR

- Spindles: amplitude rises and falls every 5–20 beats
- Polarity twists above and below the baseline
- Long QT in the sinus beats before the run
- Runs often self-terminate, then recur

ACT

- **Pulseless or sustained → defibrillate** — Unsynchronized
- **Magnesium sulfate 1–2 g IV** — In arrest: 1–2 g IV/IO push, diluted
- **Stop QT-prolonging drugs; replace K^+** — Review every medication
- **× Avoid QT-prolonging antiarrhythmics**

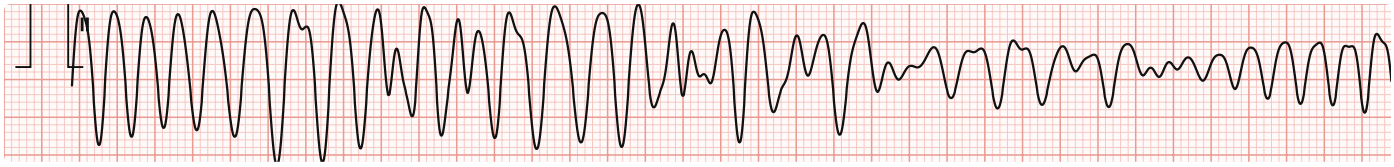
Don't confuse with ≠ **Polymorphic VT** normal baseline QT ≠ **VF** no organised spindle pattern

CARDIAC ARREST

7 rhythms · 25 mm/s, 10 mm/mV

Coarse ventricular fibrillation ARREST

Chaotic, disorganised ventricular activity with large waves — no cardiac output.



CRITERIA

Rate	Not measurable (waves ~300–600/min)
Rhythm	Chaotic
P / QRS / T	None identifiable
Amplitude	Waves > 3 mm
Pulse	Absent

LOOK FOR

- No identifiable P, QRS or T waves
- Irregular waves that vary in size and shape
- An unresponsive, pulseless patient — if the patient is talking, check the leads

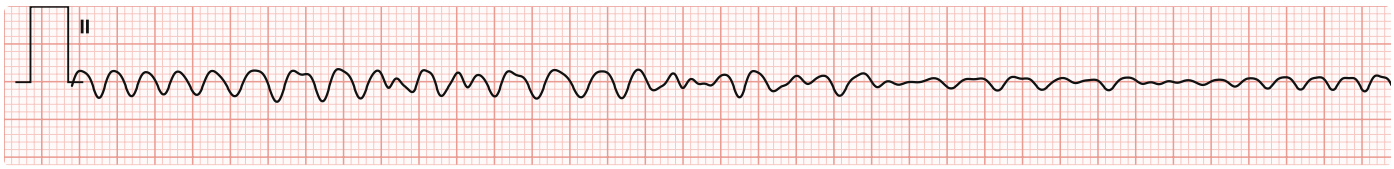
ACT

- **CPR and defibrillate** — Shock as soon as possible (biphasic per manufacturer, typically 120–200 J
- **Epinephrine 1 mg IV/IO** — Every 3–5 min — commonly after the second shock
- **Amiodarone 300 mg IV/IO** — After the third shock
- **Refractory VF** — After ≥ 3 shocks consider a vector change (anterior–posterior pads) ...

Don't confuse with ≠ **Fine VF** small waves — still shockable ≠ **Polymorphic VT** identifiable wide complexes

Fine ventricular fibrillation ARREST

Low-amplitude VF that can be mistaken for asystole.



CRITERIA

Rate	Not measurable
Rhythm	Chaotic
Amplitude	Waves < 3 mm
Pulse	Absent

LOOK FOR

- Small chaotic waves — the baseline is never truly flat
- Confirm in a second lead and increase the gain
- Treat as VF: it is shockable

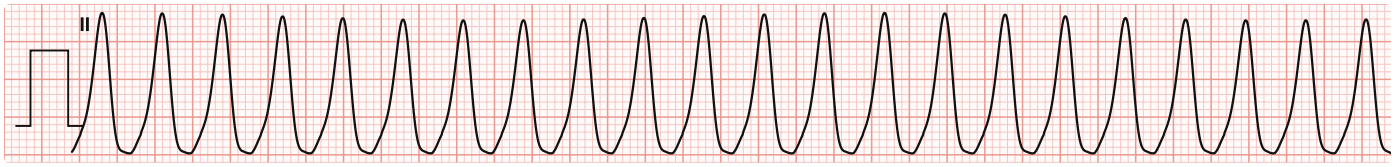
ACT

- **Defibrillate** — Fine VF is shockable — same algorithm as coarse VF
- **Confirm in another lead** — Increase the gain
- **High-quality CPR and epinephrine** — CPR 2 min between rhythm checks

Don't confuse with ≠ **Asystole** flat in two leads at high gain ≠ **Coarse VF** larger waves

Pulseless VT ARREST

Ventricular tachycardia without a pulse — treated exactly like VF.



CRITERIA

Rate	Usually > 150/min
Rhythm	Regular
QRS	Wide, identical
Pulse	Absent

LOOK FOR

- Looks like VT on the monitor
- No palpable pulse and no pleth — the diagnosis is clinical
- Manage as VF

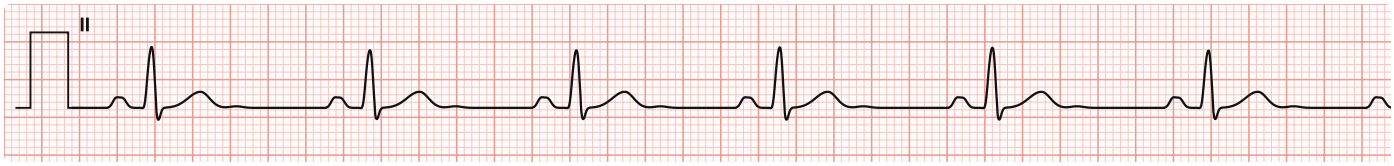
ACT

- **Defibrillate and start CPR** — Unsynchronized shock, then 2 min CPR
- **Epinephrine and amiodarone or lidocaine** — As for VF

Don't confuse with ≠ **VT with a pulse** synchronized cardioversion or drugs ≠ **VF** no organised complexes

Pulseless electrical activity ARREST

An organised rhythm on the monitor with no palpable pulse.



CRITERIA

Rate	Any (often slow)
Rhythm	Organised
QRS	Narrow or wide
Pulse	Absent — no pleth waveform

LOOK FOR

- Organised complexes on the monitor
- No pulse and a flat pleth trace
- The monitor still shows a heart rate — that is not a pulse

ACT

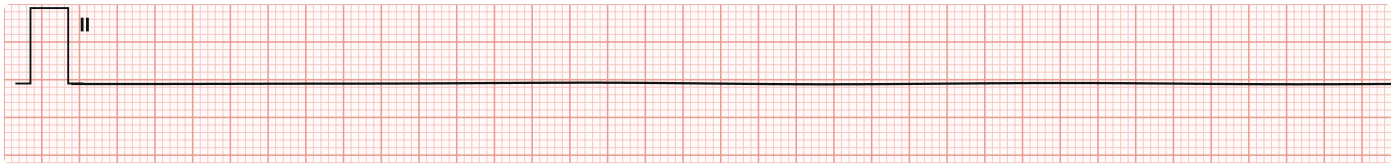
- **CPR immediately** — High-quality CPR with minimal interruptions
- **Epinephrine 1 mg IV/IO as soon as possible** — Then every 3–5 min
- **Hunt for reversible causes** — Hypovolaemia, hypoxia, H^+ (acidosis), hypo-/hyperkalaemia, hypothermia

× Do not shock

Don't confuse with ≠ **Perfusing sinus rhythm** pulse and pleth present ≠ **IVR with a pulse** bradycardia algorithm

Asystole ARREST

No ventricular electrical activity — a flat line.



CRITERIA

Rate	None
Rhythm	None
Waves	No P, QRS or T (slight drift is normal)
Pulse	Absent

LOOK FOR

- A flat line with minimal drift
- Confirm: leads attached, gain increased, second lead checked
- Fine VF can masquerade as asystole

ACT

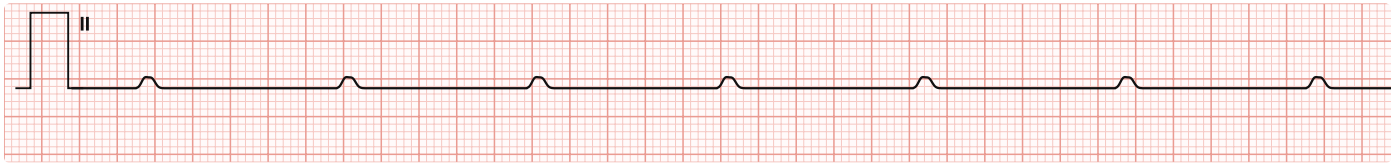
- **CPR immediately** — High-quality CPR
- **Epinephrine 1 mg IV/IO as soon as possible** — Then every 3–5 min
- **Confirm true asystole** — Check leads, raise the gain, view another lead — exclude fine VF

× Do not shock or pace

Don't confuse with ≠ **Fine VF** small chaotic waves ≠ **P-wave asystole** atrial activity only

P-wave asystole (ventricular standstill) ARREST

The atria still depolarise, but nothing reaches the ventricles.



CRITERIA

Atrial rate	Present and regular
QRS	None
Pulse	Absent

LOOK FOR

- P waves with no QRS complexes
- No pulse

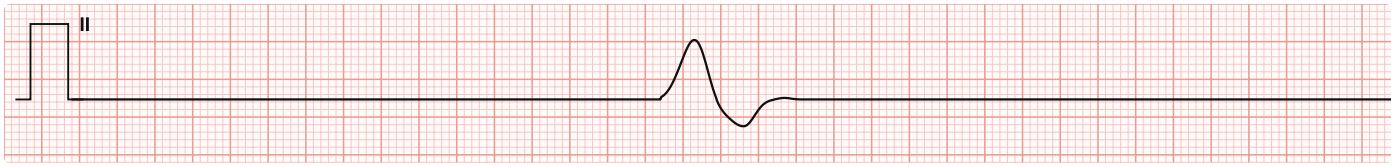
ACT

- **CPR and epinephrine** — Non-shockable arrest algorithm

Don't confuse with ≠ **Asystole** no P waves ≠ **CHB** P waves plus a slow escape rhythm

Agonal rhythm ARREST

The dying heart: very slow, very wide, irregular ventricular complexes.



CRITERIA

Rate	< 20/min
Rhythm	Irregular
QRS	Very wide, low amplitude
Pulse	Absent

LOOK FOR

- Occasional bizarre, broad complexes
- Long flat intervals between them
- No pulse

ACT

- **CPR and epinephrine** — Non-shockable (PEA) pathway
- **Reversible causes** — Especially hyperkalaemia, hypoxia and acidosis

Don't confuse with ≠ **IVR** regular, 20–40/min ≠ **Asystole** no complexes

AV BLOCK

6 rhythms · 25 mm/s, 10 mm/mV

1st-degree AV block STABLE

Every P wave conducts, but slowly: PR > 200 ms and constant.



CRITERIA

Rate	Usually normal
Rhythm	Regular
P waves	Normal, one per QRS
PR	> 200 ms, constant
QRS	Narrow

LOOK FOR

- A long, constant PR interval (more than one large box)
- Every P is followed by a QRS
- At faster rates the P approaches the preceding T

ACT

- **Usually no treatment** — Review AV-nodal drugs

Don't confuse with ≠ **Mobitz I** PR lengthens and then a beat drops ≠ **NSR** PR 120–200 ms

2nd-degree AV block, Mobitz I (Wenckebach) ASSESS

The PR lengthens beat by beat until a P wave fails to conduct — then the cycle repeats.



CRITERIA

Atrial	Regular
Ventricular	Irregular — grouped beating
P waves	Regular; periodically not conducted
PR	Progressively longer; shortest after the pause
QRS	Usually narrow

LOOK FOR

- Grouped beating with regular pauses
- Longest PR just before the dropped QRS, shortest just after
- R–R shortens through the group; the pause is less than two P–P intervals

ACT

- **Symptomatic → atropine 1 mg IV** — Usually effective at AV-nodal level

Don't confuse with ≠ **Mobitz II** PR constant before the dropped beat ≠ **2:1 block** type cannot be determined

2nd-degree AV block, Mobitz II URGENT

Constant PR — then a P wave suddenly fails to conduct.



CRITERIA

Atrial	Regular
Ventricular	Irregular
P waves	Regular; intermittently not conducted
PR	Constant in conducted beats
QRS	Often wide (infranodal; here RBBB)

LOOK FOR

- Constant PR before and after the dropped beat
- The pause is exactly twice the P–P interval
- Often accompanied by bundle branch block

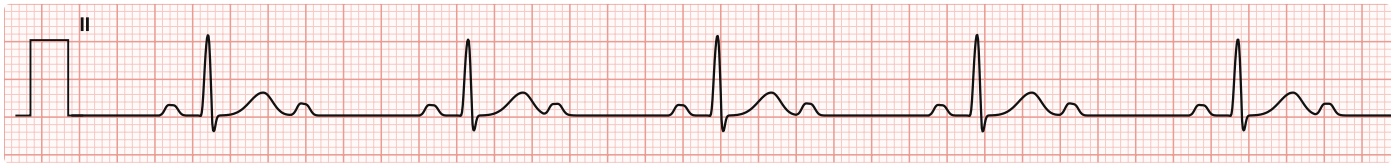
ACT

- **Prepare transcutaneous pacing** — Pads on
- **Epinephrine 2–10 mcg/min or dopamine 5–20 mcg/kg/min** — Bridge if pacing is unavailable or ineffective
- **Expert consultation** — Transvenous then permanent pacing

Don't confuse with ≠ **Mobitz I** PR lengthens ≠ **2:1 block** 2:1 block

2:1 AV block URGENT

Every second P wave is blocked — Mobitz I and II cannot be told apart on this strip.



CRITERIA

Atrial	Regular
Ventricular	Half the atrial rate
P waves	Two for every QRS (one may hide in the T)
PR	Constant in conducted beats
QRS	Narrow suggests nodal; wide suggests infranodal

LOOK FOR

- Two P waves for every QRS — find the one hiding near the T wave
- Constant PR in conducted beats
- Longer strips may reveal Wenckebach periods

ACT

- **Assess perfusion** — Symptomatic → bradycardia algorithm
- **Pads on, prepare to pace** — Especially if the QRS is wide
- **Atropine 1 mg IV** — May help if the block is nodal (narrow QRS)

Don't confuse with ≠ **Sinus bradycardia** one P per QRS — look for a hidden P ≠ **Mobitz I** Mobitz I

High-grade (advanced) AV block URGENT

Two or more consecutive P waves fail to conduct (here 3:1).



CRITERIA

Atrial	Regular
Ventricular	Slow
P waves	≥ 2 consecutive non-conducted
PR	Constant in conducted beats
QRS	Narrow or wide

LOOK FOR

- Several P waves in a row without a QRS
- Conducted beats keep a constant PR
- Very slow ventricular rate

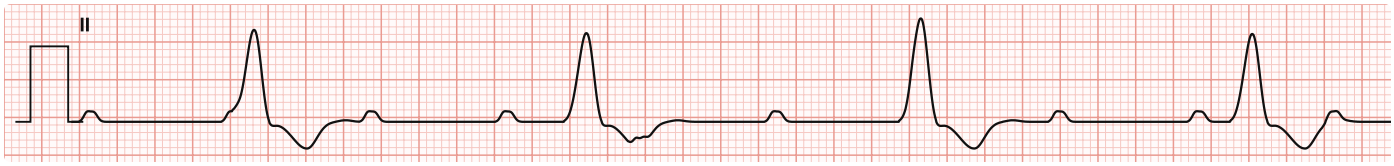
ACT

- **Transcutaneous pacing** — If symptomatic or poorly perfused
- **Epinephrine or dopamine infusion** — Bridge to pacing

Don't confuse with ≠ **CHB** no consistent PR at all ≠ **2:1 block** one blocked P at a time

3rd-degree (complete) heart block URGENT

No atrial impulses reach the ventricles: P waves and QRS complexes march independently.



CRITERIA

Atrial	Regular
Ventricular	Regular and slower (wide escape 20–40, junctional 40–60)
P waves	More P waves than QRS, unrelated
PR	Varies with no pattern
QRS	Wide (ventricular) or narrow (junctional) escape

LOOK FOR

- Regular P–P and regular R–R at different rates
- PR varies randomly — P waves hide in QRS complexes and T waves
- Escape QRS narrow (junctional) or wide (ventricular)

ACT

- **Transcutaneous pacing** — First-line if symptomatic — confirm electrical AND mechanical capture
- **Epinephrine 2–10 mcg/min or dopamine 5–20 mcg/kg/min** — If pacing is delayed or ineffective
- **Expert consultation** — Transvenous pacing — TCP is a bridge

Don't confuse with ≠ **2:1 block** fixed PR in conducted beats ≠ **IVR** no P waves

CONDUCTION

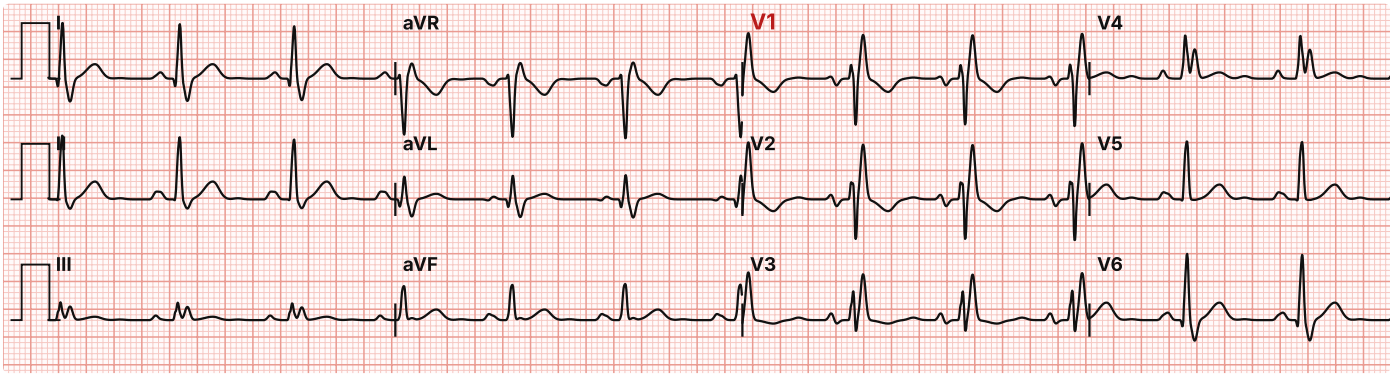
4 rhythms · 12-lead at 74%

Right bundle branch block

STABLE

Look at V1

Late right-ventricular activation: rSR′ in V1 and a broad S in I and V6.



CRITERIA

QRS	≥ 120 ms
V1–V2	rSR′ ("M" shape)
I, aVL, V5–V6	Wide, slurred S wave
ST–T	Secondary T inversion in V1–V3

LOOK FOR

- rSR′ "rabbit ears" in V1
- Wide S wave in lead I and V6
- T inversion in V1–V3 is expected (secondary)

ACT

- **Compare with previous ECGs** — New RBBB with chest pain or breathlessness: consider ACS and PE

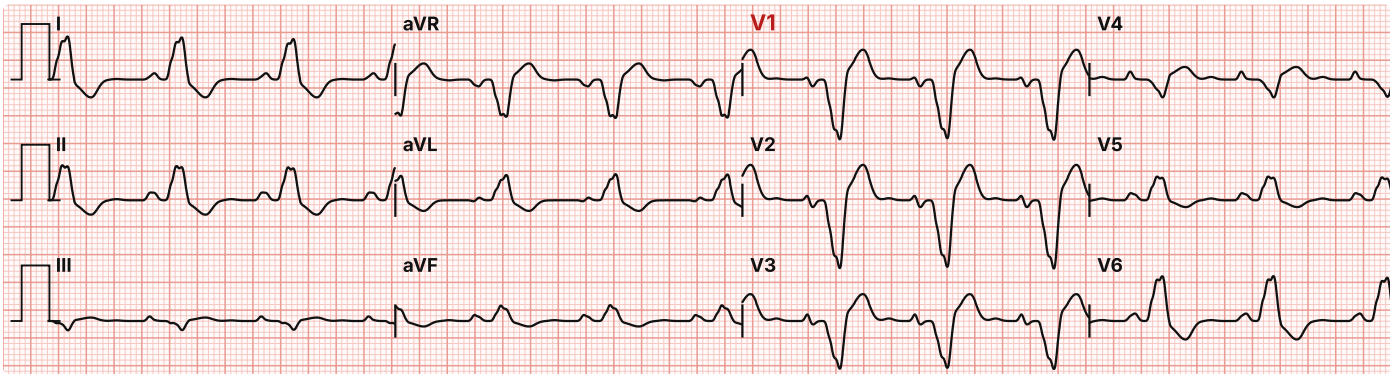
Don't confuse with ≠ **LBBB** broad notched R in V6, QS in V1 ≠ **WPW** short PR and delta wave

Left bundle branch block

ASSESS

Look at V1

Late left-ventricular activation: deep QS in V1 and a broad notched R in I and V6.



CRITERIA

QRS	≥ 120 ms
V1	QS or rS
I, aVL, V5–V6	Broad notched R without a septal q
ST–T	Discordant to the QRS

LOOK FOR

- Broad notched ("M") R in I and V6 — no septal q
- Deep QS in V1–V3
- ST and T point opposite to the main QRS (appropriate discordance)

ACT

- **Ischaemic symptoms → apply (modified) Sgarbossa criteria** — Positive criteria → treat as a STEMI equivalent and discuss with cardiology

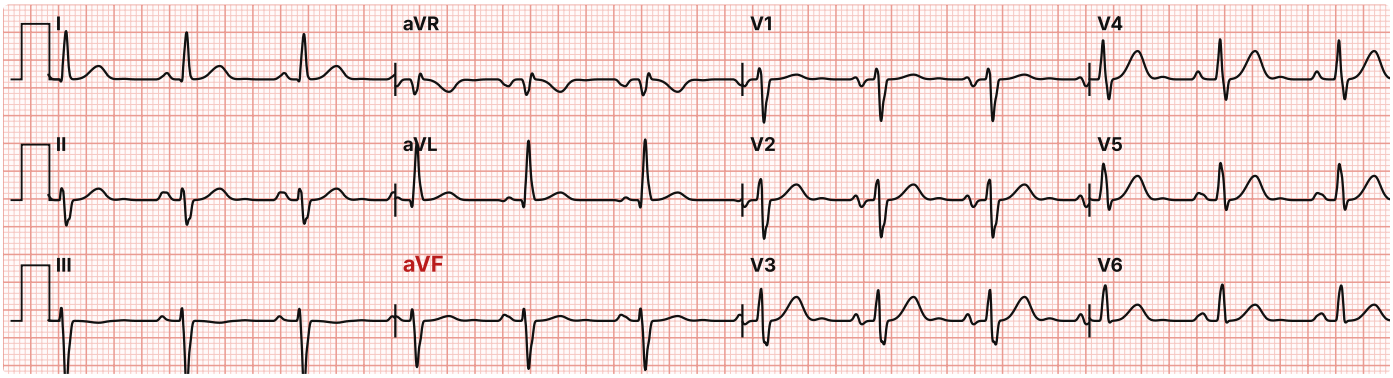
Don't confuse with ≠ **RBBB** rSR′ in V1 ≠ **Ventricular pacing** a spike before each QRS

Left anterior fascicular block

STABLE

Look at aVF

Block of the left anterior fascicle: marked left axis deviation with a near-normal QRS.



CRITERIA

Axis	−45° to −90°
I, aVL	qR
II, III, aVF	rS
QRS	< 120 ms

LOOK FOR

- Lead I positive; II and aVF negative
- Small q with a tall R in aVL
- QRS only slightly widened

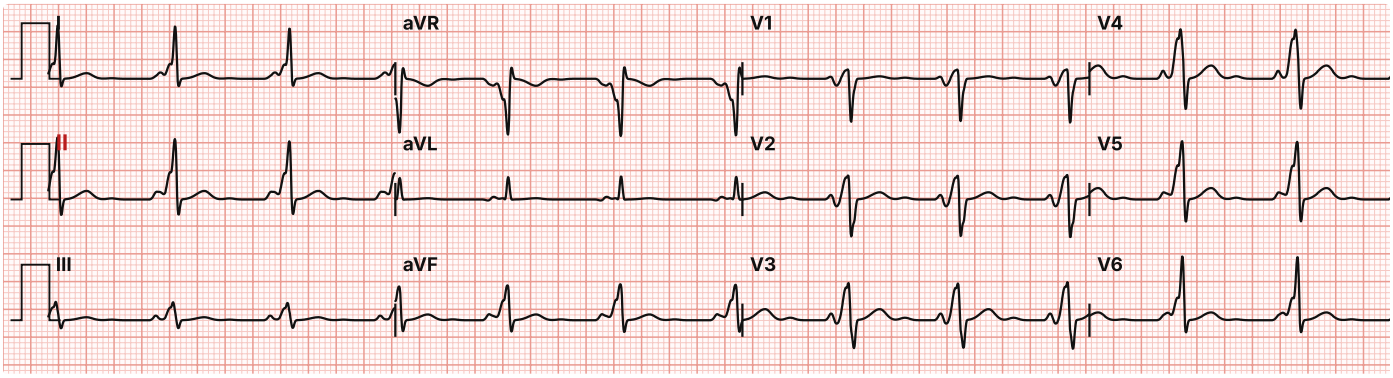
ACT

- **No specific treatment** — Look for associated RBBB or AV block

Don't confuse with ≠ **RV pacing** superior axis but wide QRS with spikes ≠ **Old inferior MI** Q waves (not rS) in II, III, aVF

Wolff–Parkinson–White pattern ASSESS

Ventricular pre-excitation: short PR, delta wave and a wide QRS.



CRITERIA

PR	< 120 ms
Delta wave	Slurred QRS upstroke
QRS	> 110–120 ms
ST–T	Secondary changes

LOOK FOR

- Short PR with no flat PR segment
- Slurred upstroke of the QRS (delta wave)
- Pseudo-infarct Q waves may appear in some leads

ACT

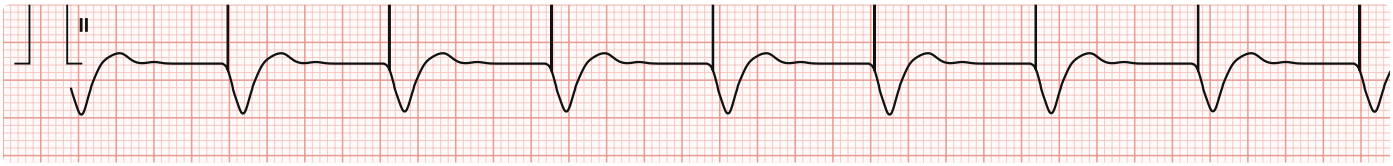
- **Electrophysiology referral** — Especially with palpitations or syncope
- **Tachycardia with WPW** — Narrow orthodromic AVRT can be managed like SVT

Don't confuse with ≠ **Bundle branch block** normal PR ≠ **Pre-excited AF** the dangerous complication

PACEMAKER 4 rhythms · 25 mm/s, 10 mm/mV

Ventricular paced rhythm (VVI) STABLE

A pacing spike before every wide QRS — the device is driving the ventricle.



CRITERIA

Rate	Programmed (here 70/min)
Rhythm	Regular
Spikes	Before each QRS
QRS	Wide, LBBB-like, superior axis (RV apex)
T	Discordant

LOOK FOR

- Spike immediately followed by a wide QRS (capture)
- Negative QRS in II, III, aVF with an RV apical lead
- Constant spike-to-spike interval

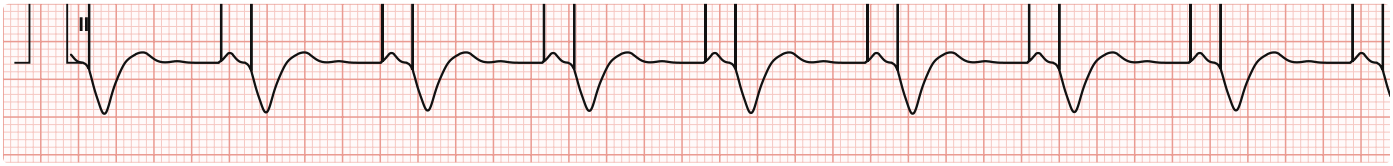
ACT

- **Confirm capture and a pulse** — Every spike followed by a QRS and a palpable pulse

Don't confuse with ≠ **LBBB** no spikes, P waves conduct ≠ **Failure to capture** spikes without QRS

Dual-chamber paced rhythm (DDD) STABLE

Atrial spike → paced P wave; ventricular spike → paced QRS.



CRITERIA

Spikes	Two per cycle
AV delay	~120–200 ms
QRS	Wide paced complex

LOOK FOR

- Two spikes per cycle
- Small paced P after the first spike
- Wide paced QRS after the second

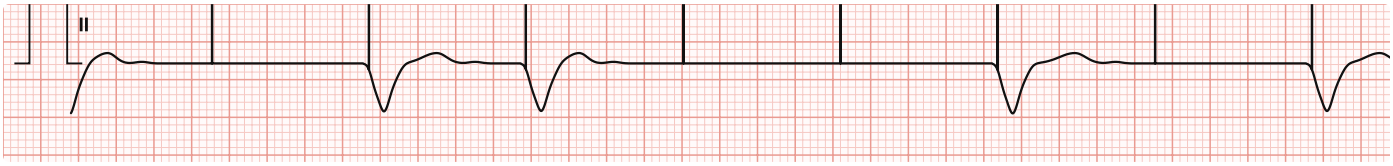
ACT

- **Confirm capture in both chambers** — Each spike should produce a P wave or QRS, and a pulse

Don't confuse with ≠ **VVI** ventricular spike only ≠ **LBBB** no spikes

Pacemaker failure to capture URGENT

Pacing spikes appear, but some are not followed by a QRS.



CRITERIA

Spikes	At the programmed rate
Capture	Some spikes without a QRS
Result	Pauses and bradycardia

LOOK FOR

- A spike with no QRS after it
- Pauses and a slow ventricular rate
- The spike falls outside the refractory period — true non-capture

ACT

- **Treat symptomatic bradycardia** — Transcutaneous pacing
- **Correct causes** — Hyperkalaemia, acidosis, ischaemia
- **Urgent device interrogation** — Output can be increased with the programmer

Don't confuse with ≠ **Undersensing** spikes in the wrong place ≠ **Normal pacing** every spike captures

Pacemaker undersensing URGENT

The pacemaker ignores the patient's own beats and fires anyway — spikes land on QRS complexes and T waves.



CRITERIA

Intrinsic rhythm	Present
Spikes	Fixed rate, unrelated to native beats
Capture	Only when the ventricle is not refractory

LOOK FOR

- Spikes too soon after native beats
- Spikes landing on T waves (R-on-T risk)
- A mixture of paced and native beats

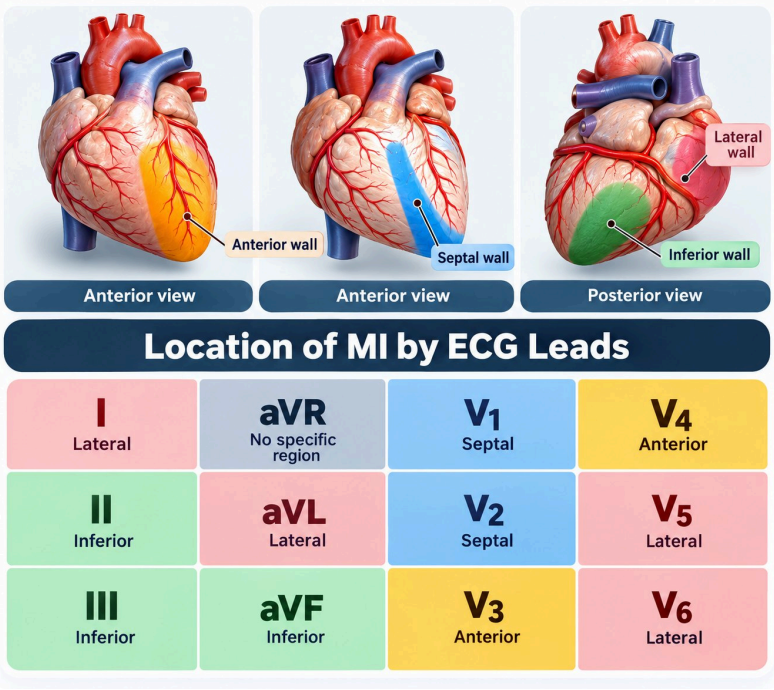
ACT

- **Monitor closely with pads available** — R-on-T risk
- **Device interrogation** — Adjust sensitivity

Don't confuse with ≠ **Failure to capture** spikes without QRS ≠ **V-paced** Normal VVI pacing

ACS & ISCHAEMIA

11 rhythms · 12-lead at 74%



Where is the infarct?

ST elevation in two or more contiguous leads localises the injured wall; the opposite wall shows reciprocal depression.

WALL	LEADS	USUAL ARTERY
Inferior	II, III, aVF	RCA (~80%) or LCx
Septal	V1–V2	LAD — septal branches
Anterior	V3–V4	LAD
Lateral	I, aVL, V5–V6	LCx or diagonal
Posterior	ST ↓ V1–V3 · STE V7–V9	RCA or LCx
Right ventricle	V1, V4R	Proximal RCA

aVR looks at no single wall. STE in aVR with diffuse ST depression suggests left-main or three-vessel disease.

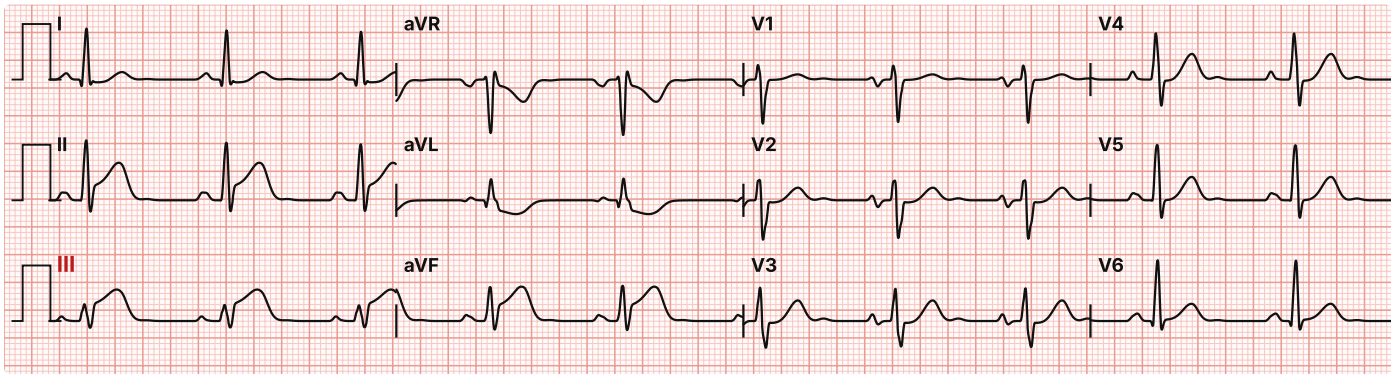
Inferior STEMI? Record V4R before nitrates (RV involvement) and V7–V9 if V1–V3 are depressed.

Reciprocal pairs Inferior ↔ I/aVL · Anterior ↔ inferior · Posterior ↔ V1–V3.

Inferior STEMI URGENT

Look at III

ST elevation in II, III and aVF with reciprocal depression in I and aVL — usually the right coronary artery.



CRITERIA

Rate	Often slow (vagal)
ST elevation	II, III, aVF (III > II suggests RCA)
Reciprocal	ST depression in I and aVL
Also check	V4R for RV involvement; AV block

LOOK FOR

- ST elevation in II, III and aVF
- Reciprocal ST depression in aVL — often the earliest clue
- STE in III greater than II points to the RCA: check V1 and V4R

ACT

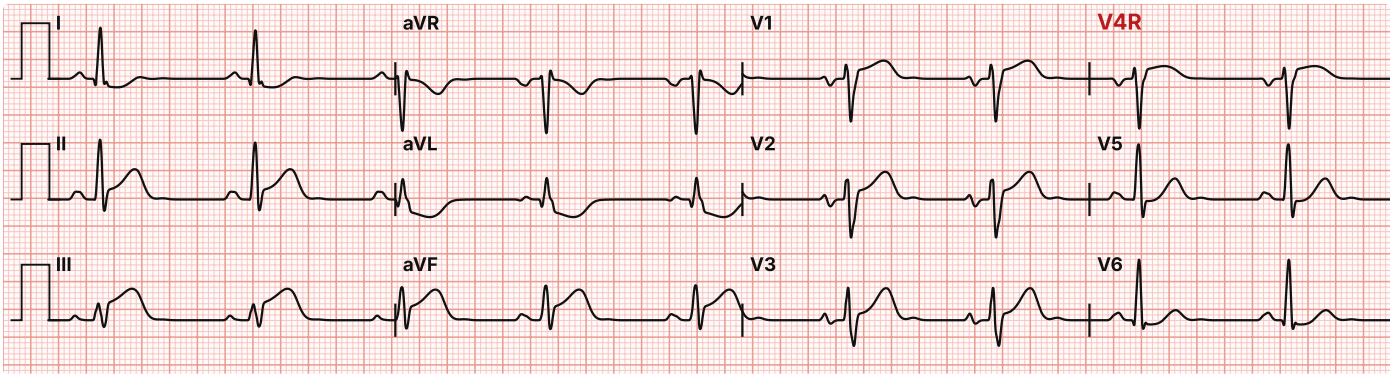
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%
- **Right-sided leads before nitrates** — STE in V4R → avoid nitrates
- **Nitroglycerin 0.4 mg SL — with caution** — Only if SBP ≥ 90 mmHg, no RV infarction and no PDE-5 inhibitor use

Don't confuse with ≠ **Inferior + RV** STE also in V1 and V4R ≠ **Pericarditis** diffuse STE, PR depression, no reciprocal change except aVR

Inferior STEMI with RV involvement URGENT

Look at V4R

Inferior STEMI plus ST elevation in V1 and right-sided V4R — the right ventricle is infarcted.



CRITERIA

ST elevation	II, III, aVF (III > II), V1, V4R ≥ 1 mm
Reciprocal	I, aVL
Clinical	Hypotension with clear lung fields

LOOK FOR

- STE in III more than II
- STE in V1 — unusual for a pure inferior MI
- STE ≥ 1 mm in V4R confirms RV involvement

ACT

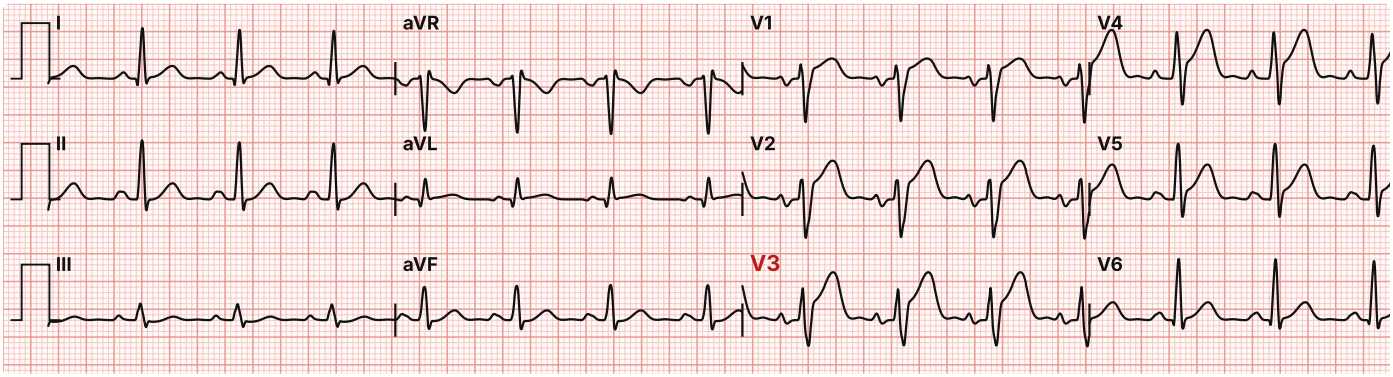
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%
- **IV fluid bolus for hypotension** — With clear lungs
- **× Avoid nitrates**

Don't confuse with ≠ **Inferior STEMI without RV** V1 and V4R not elevated ≠ **Anteroseptal STEMI** STE V1–V3 without inferior STE

Anterior STEMI URGENT

Look at V3

ST elevation in V2–V4 — occlusion of the left anterior descending artery.



CRITERIA

ST elevation	V2–V4 (V2–V3: ≥ 2 mm men ≥ 40 y, ≥ 2.5 mm men < 40 y, ≥ 1.5 mm women; ≥ 1 mm other leads)
Reciprocal	II, III, aVF
Evolution	Q waves and T inversion over hours

LOOK FOR

- ST elevation across V2–V4
- Hyperacute, broad-based T waves early
- Reciprocal depression in II, III, aVF

ACT

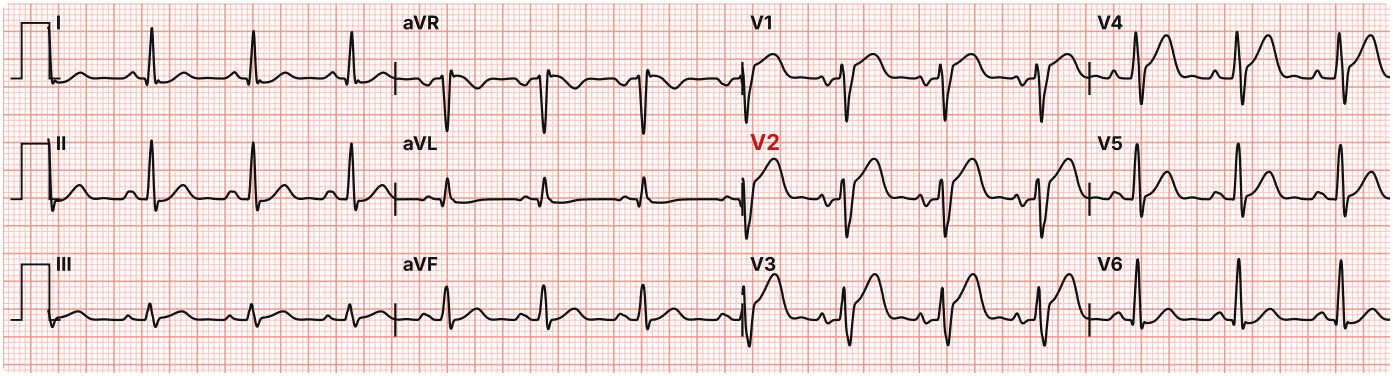
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%

Don't confuse with ≠ **Septal** maximal in V1–V2 ≠ **Proximal LAD** V1–V6, I, aVL

Septal / anteroseptal STEMI URGENT

Look at V2

ST elevation in V1–V3 — the LAD and its septal branches.



CRITERIA

ST elevation	V1–V2 (± V3)
Watch for	New RBBB, LAFB or AV block

LOOK FOR

- STE in V1 and V2 (± V3)
- New bundle branch block or AV block
- Often extends into a larger anterior infarct

ACT

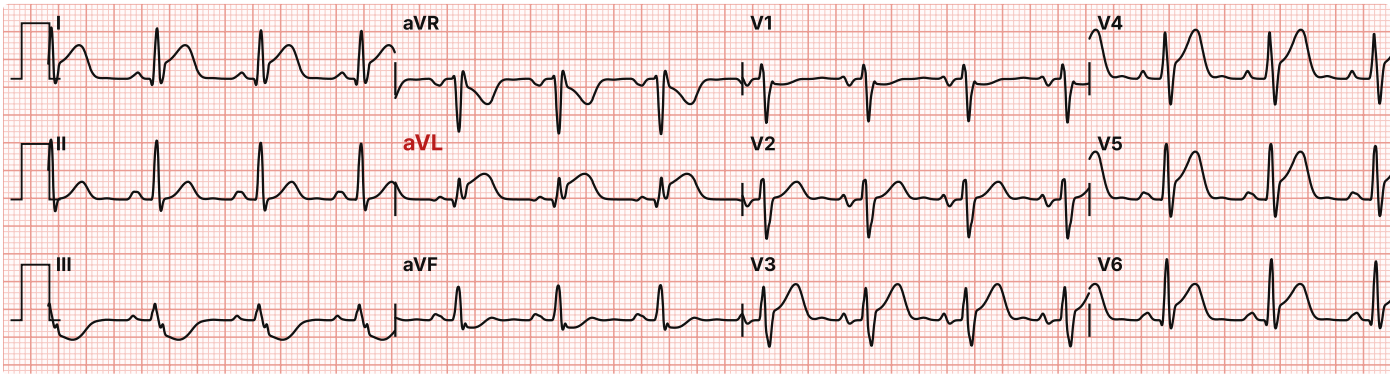
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%
- **Monitor for conduction block** — Pads on

Don't confuse with ≠ **Anterior** V2–V4 ≠ **Brugada** coved STE in V1–V2 without evolution or reciprocal change

Lateral STEMI URGENT

Look at aVL

ST elevation in I, aVL, V5 and V6 — circumflex or diagonal artery.



CRITERIA

ST elevation	I, aVL, V5–V6
Reciprocal	III, aVF

LOOK FOR

- STE in I and aVL (high lateral) and V5–V6
- Reciprocal depression in III and aVF
- Easily missed on a single monitoring lead — always record a 12-lead

ACT

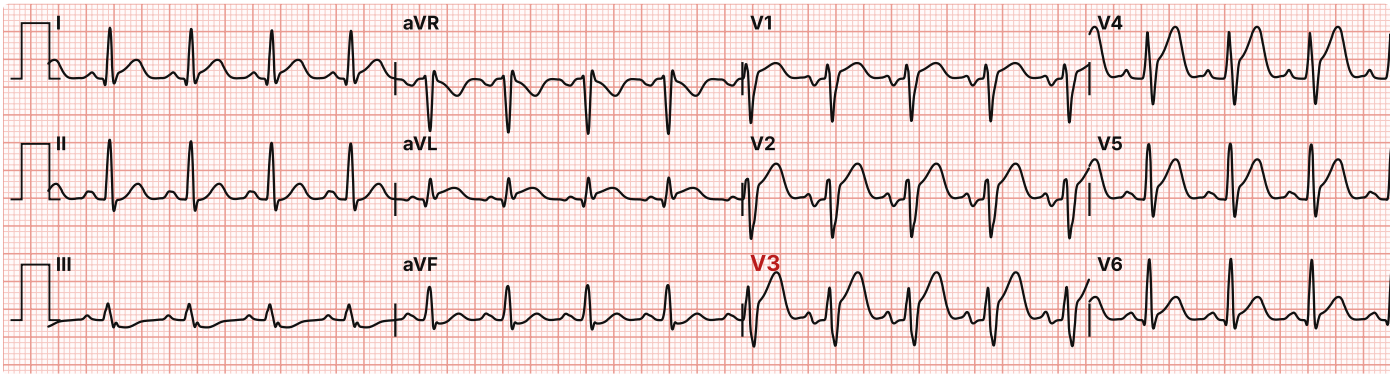
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%

Don't confuse with ≠ **Proximal LAD** STE also in V1–V4 ≠ **Pericarditis** diffuse STE without reciprocal depression

Extensive anterior STEMI (proximal LAD) URGENT

Look at V3

ST elevation in V1–V6 plus I and aVL — the "widow-maker" proximal LAD occlusion.



CRITERIA

ST elevation	V1–V6, I, aVL
Reciprocal	II, III, aVF
Rate	Often sinus tachycardia

LOOK FOR

- STE across the whole precordium and the high lateral leads
- Deep reciprocal depression inferiorly
- Sinus tachycardia suggests a large infarct or poor output

ACT

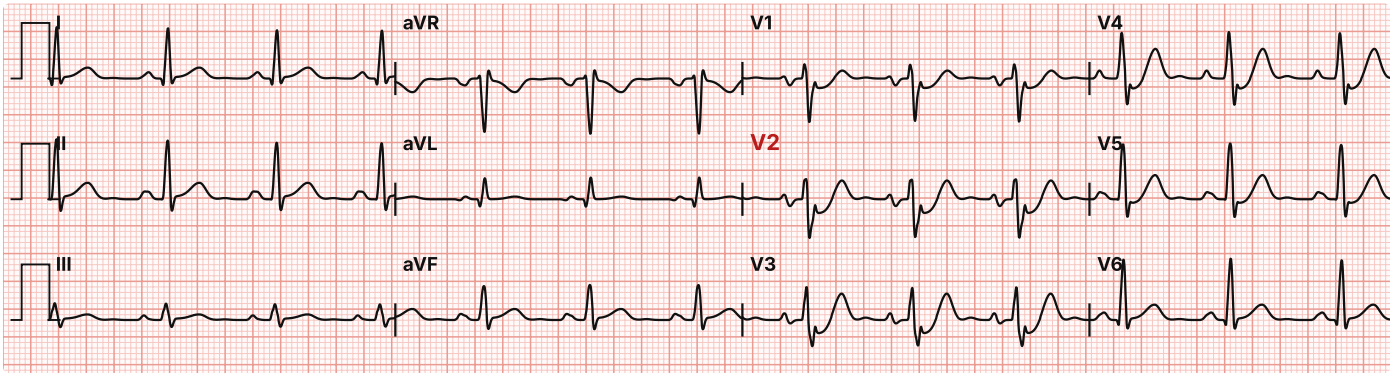
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%
- **Anticipate shock and VF** — Pads on

Don't confuse with ≠ **Mid LAD** V2–V4 ≠ **Lateral STEMI** Lateral only

Posterior STEMI URGENT

Look at V2

Horizontal ST depression in V1–V3 — the mirror image of posterior ST elevation. Confirm in V7–V9.



CRITERIA

V1–V3	Horizontal ST depression with upright T
Evolution	Tall R in V2 (R/S > 1)
V7–V9	ST elevation ≥ 0.5 mm

LOOK FOR

- ST depression maximal in V1–V3 — think posterior, not just ischaemia
- Upright T waves in V1–V3
- Record V7–V9 to see the elevation

ACT

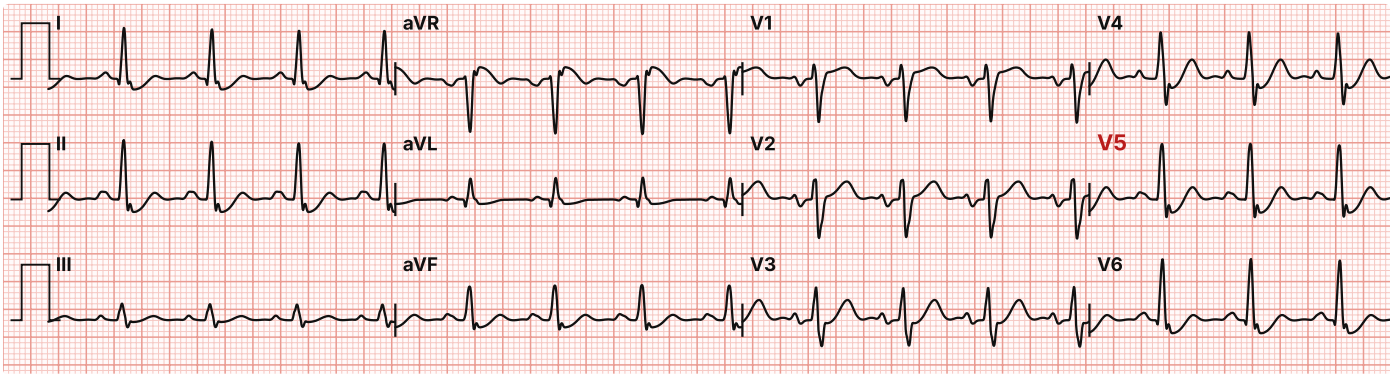
- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%
- **Record posterior leads V7–V9** — STE ≥ 0.5 mm confirms posterior STEMI

Don't confuse with ≠ **Subendocardial ischaemia** diffuse depression with STE in aVR ≠ **RBBB** tall R' in V1 with secondary T inversion

Diffuse subendocardial ischaemia (STE in aVR) URGENT

Look at V5

Widespread ST depression with ST elevation in aVR — left main or three-vessel disease, or severe supply–demand mismatch.



CRITERIA

ST depression	≥ 1 mm in ≥ 6–8 leads, maximal in I, II, V4–V6
ST elevation	aVR (± V1), aVR > V1
Rate	Often tachycardic

LOOK FOR

- ST depression in many leads, maximal in V4–V6
- ST elevation in aVR
- Horizontal or down-sloping depression

ACT

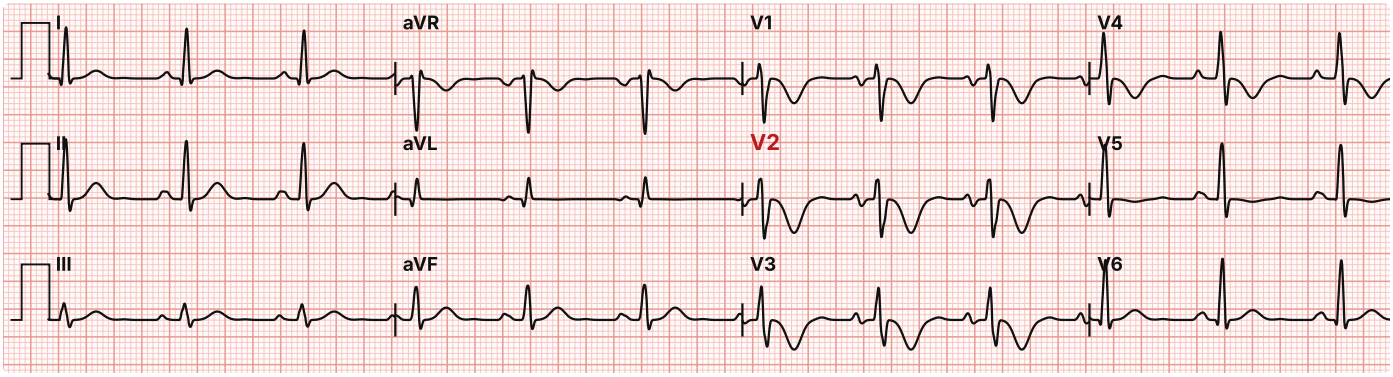
- **High-risk NSTE-ACS pathway** — Urgent cardiology review
- **Aspirin 162–325 mg chewed** — Unless true allergy or active bleeding
- **Treat precipitants** — Anaemia, hypotension, hypoxia, tachyarrhythmia, sepsis
- **Nitroglycerin for ongoing pain** — If SBP ≥ 90 mmHg and not contraindicated

Don't confuse with ≠ **Posterior STEMI** depression maximal in V1–V3 ≠ **LVH strain** lateral depression with very tall voltages

Wellens syndrome URGENT

Look at V2

Deep or biphasic T waves in V2–V3 in a pain-free patient — a critical proximal LAD stenosis.



CRITERIA

Type A (≈25%)	Biphasic T (up, then down) in V2–V3
Type B (≈75%)	Deep symmetric T inversion in V2–V3
ST	Isoelectric or < 1 mm elevation
QRS	No Q waves; R-wave progression preserved
History	Recent angina, often pain-free now

LOOK FOR

- Symmetric deep T inversion (B) or biphasic T (A) in V2–V3 — switch with the Pattern control
- No significant ST elevation and no Q waves
- The patient is often pain-free when the ECG is taken

ACT

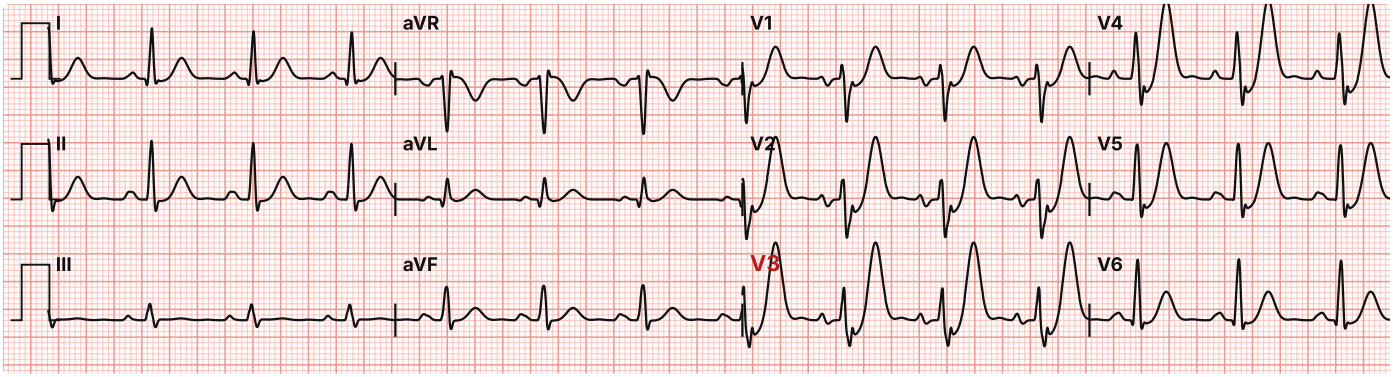
- **Urgent cardiology / angiography** — Treat as high-risk ACS even with a normal troponin
- **Aspirin and the ACS pathway** — Aspirin 162–325 mg chewed
- ✕ **Avoid stress testing**

Don't confuse with ≠ **PE** T inversion V1–V4 with tachycardia and right-heart strain ≠ **LVH strain** lateral, asymmetric inversion with high voltage

de Winter T waves URGENT

Look at V3

Up-sloping ST depression at the J point running into tall, symmetric precordial T waves — an anterior STEMI equivalent.



CRITERIA

V1–V6	1–3 mm up-sloping ST depression at the J point
T waves	Tall, prominent, symmetric
aVR	Often 0.5–1 mm STE

LOOK FOR

- J-point depression with an up-sloping ST into a tall T
- Tall, symmetric precordial T waves
- No classic ST elevation

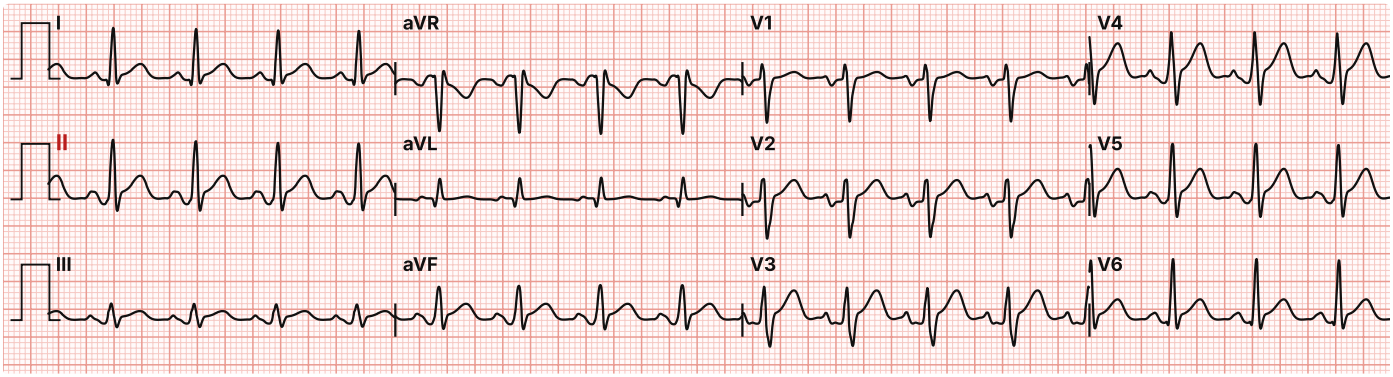
ACT

- **STEMI bundle** — Cath-lab activation, aspirin, pads on, NTG if SBP ≥ 90, O_2 only if SpO_2 < 90%

Don't confuse with ≠ **Hyperkalaemia** narrow-based peaked T without J-point depression ≠ **Anterior STEMI** ST elevation

Acute pericarditis ASSESS

Diffuse concave ST elevation with PR depression — inflammation of the whole pericardium.



CRITERIA

ST elevation	Diffuse, concave (I, II, aVF, V2–V6)
PR	Depressed in II; elevated in aVR
Reciprocal	None, except aVR (± V1)
Rate	Often tachycardic
Q waves	None

LOOK FOR

- ST elevation in many leads that do not match one coronary territory
- Concave ("saddle") ST shape
- PR depression in II and PR elevation in aVR

ACT

- **Exclude STEMI first** — A territorial pattern, reciprocal depression, convex ST or Q waves suggest STEMI
- **Echocardiography** — Check for effusion and tamponade

Don't confuse with ≠ **Inferior STEMI** territorial with reciprocal depression in aVL ≠ **Proximal LAD STEMI** convex STE with inferior reciprocal depression

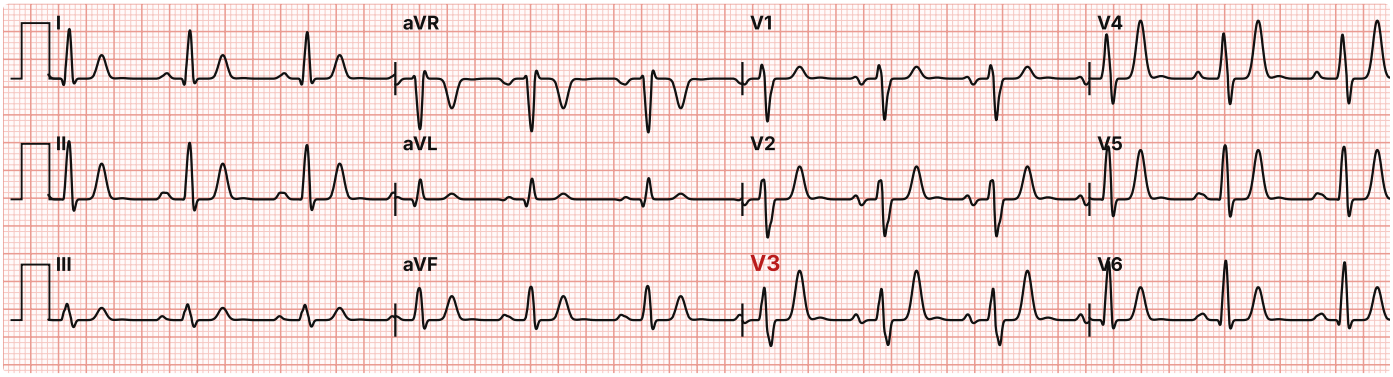
ELECTROLYTES & OTHER

11 rhythms · 12-lead at 74%

Hyperkalaemia URGENT

Look at V3

Peaked T waves → loss of P waves → wide QRS → sine wave as potassium rises. Drag the K⁺ control to watch it progress.



CRITERIA

5.5–6.5 mmol/L	Tall, narrow-based peaked T waves
6.5–8 mmol/L	Long PR, flattened P, widening QRS
> 8 mmol/L	Absent P, very wide QRS merging with T, sine wave, VF or asystole

LOOK FOR

- Tall, narrow-based, symmetric ("tented") T waves
- Flattened or absent P waves with a long PR
- Progressive QRS widening that merges into the T

ACT

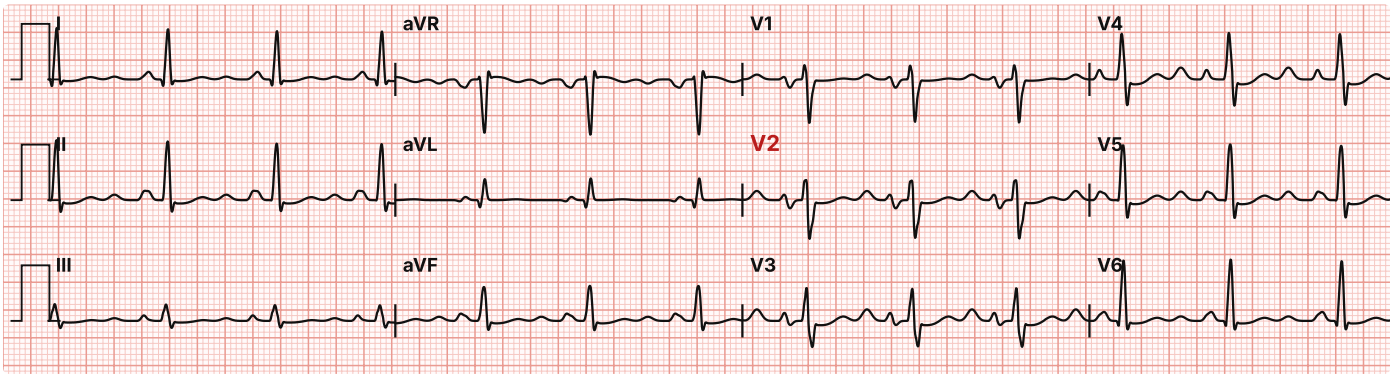
- **Calcium to stabilise the myocardium** — Calcium gluconate 10% 30 mL (3 g) IV, or calcium chloride 10% 10 mL (1 g) via a large vein
- **Shift K^+ into cells** — Insulin 10 units IV with dextrose 25 g
- **Remove K^+** — Stop K^+-raising drugs

Don't confuse with ≠ **de Winter** J-point depression with tall T waves ≠ **Hyperacute T of early STEMI** broad-based, territorial

Hypokalaemia ASSESS

Look at V2

Flattened T waves, prominent U waves and ST depression.



CRITERIA

T waves	Flat or inverted
U waves	Prominent (> 1 mm), best in V2–V3
ST	Depressed
QT	Apparently long (really QU)

LOOK FOR

- A prominent U wave after the T, best in V2–V3
- Low, flat T waves
- Mild ST depression

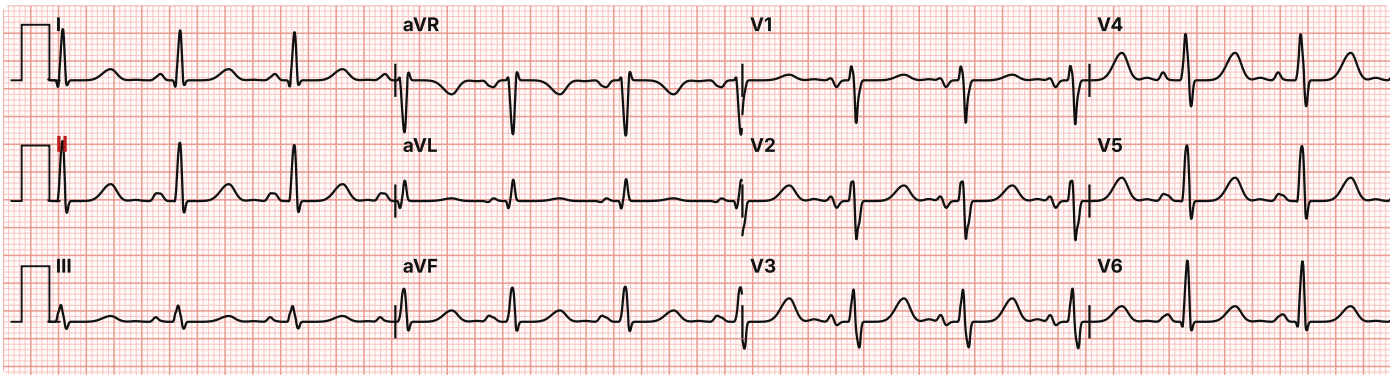
ACT

- **Replace potassium** — Oral or IV per protocol
- **Replace magnesium too** — Hypokalaemia is refractory until Mg^{2+} is corrected

Don't confuse with ≠ **Long QT** long QT without a separate U wave ≠ **Digoxin effect** scooped ST, short QT

Hypocalcaemia ASSESS

A long QT made by a long, flat ST segment — the T wave itself looks normal.



CRITERIA

QTc	Prolonged
ST segment	Lengthened and flat
T wave	Normal shape, pushed late

LOOK FOR

- A long, straight ST segment
- A normal-looking T wave arriving late
- Prolonged QTc in the measurements panel

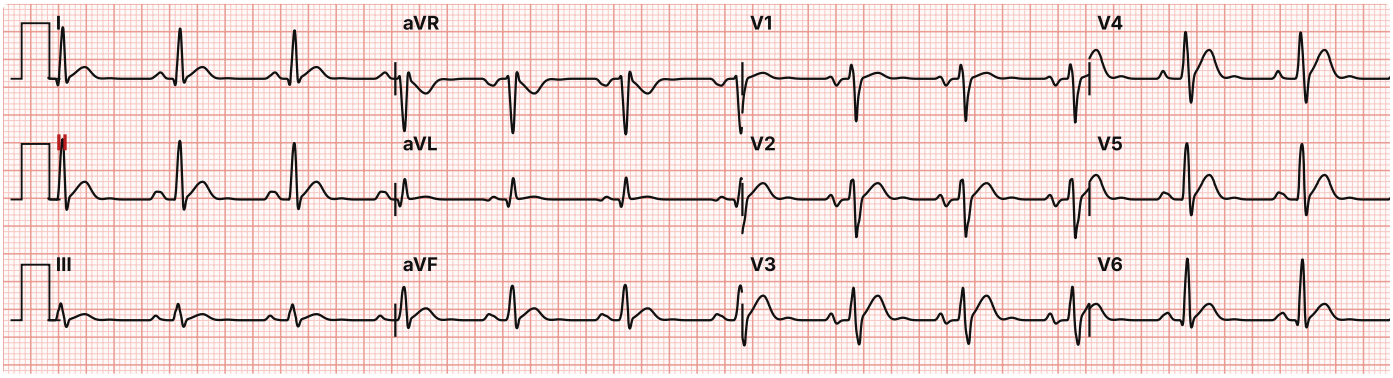
ACT

- **IV calcium if symptomatic** — Calcium gluconate 10% 10–20 mL IV over 10 min, then an infusion per protocol

Don't confuse with ≠ **Long QT syndrome** broad T wave rather than a long ST ≠ **Hypercalcaemia** short QT

Hypercalcaemia ASSESS

A short QT — the T wave starts almost straight after the QRS.



CRITERIA

QTc	Short (< 360 ms)
ST segment	Short or absent
Severe	J waves, bradyarrhythmias, AV block

LOOK FOR

- A short or absent ST segment
- Short QT interval

ACT

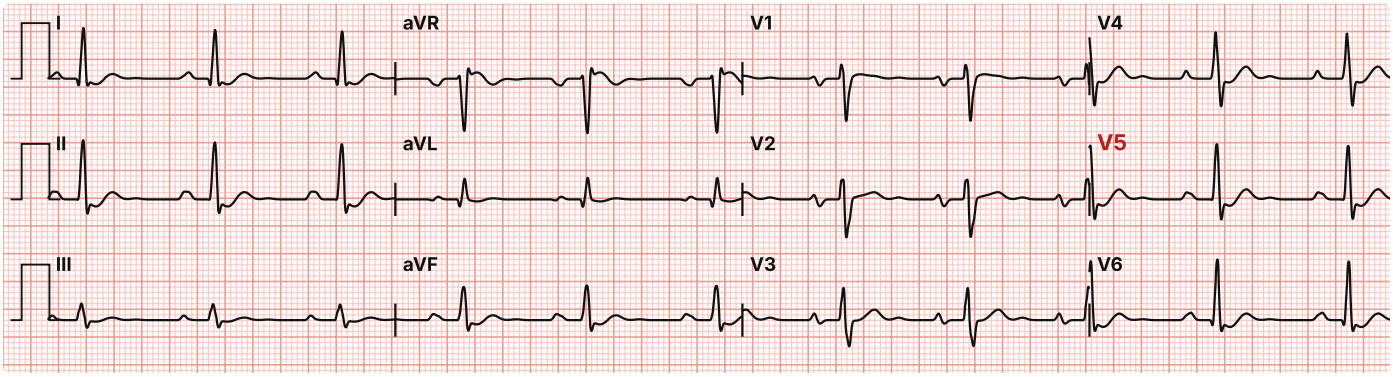
- **IV fluids and treat the cause** — Isotonic saline

Don't confuse with ≠ **Hypocalcaemia** long ST, long QT ≠ **Digoxin** short QT with scooped ST

Digoxin effect ASSESS

Look at V5

Scooped, sagging ST depression with a short QT — a drug effect, not toxicity by itself.



CRITERIA

ST	Down-sloping, scooped ("reverse tick") in leads with tall R waves
T	Flattened or inverted
QT	Short
PR	May be prolonged

LOOK FOR

- Sagging ST depression with a curved, scooped shape
- Short QT
- Mildly long PR

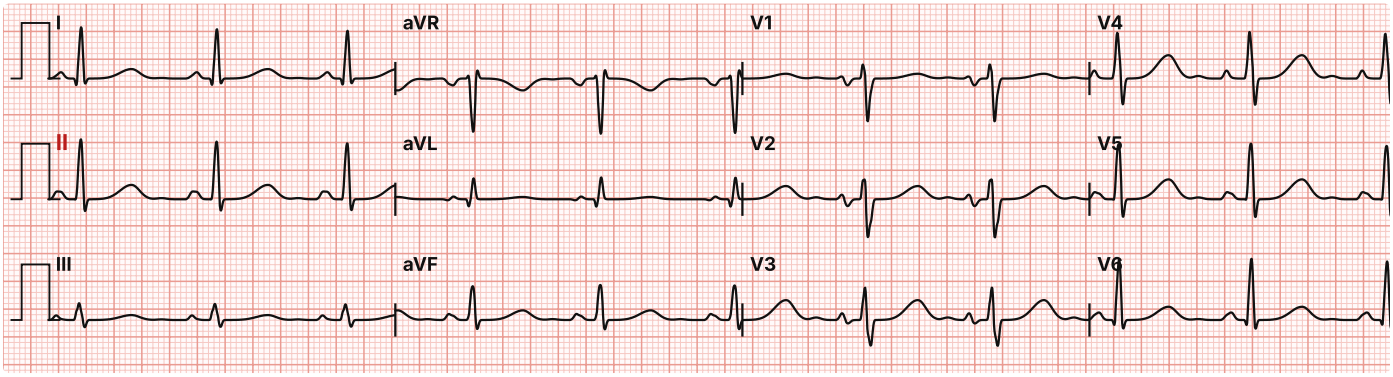
ACT

- **Correlate with level, K^+ and renal function** — Effect alone needs no treatment
- **Toxic arrhythmia → digoxin-specific antibody fragments** — For life-threatening arrhythmias or severe hyperkalaemia

Don't confuse with ≠ **Ischaemia** horizontal or down-sloping, not scooped ≠ **LVH strain** asymmetric T inversion with high voltage

Long QT URGENT

Prolonged ventricular repolarization: QTc ≥ 470 ms (men) or ≥ 480 ms (women); > 500 ms is high risk.



CRITERIA

QTc	> 460–480 ms
T wave	Broad, sometimes notched
Risk	Torsades de pointes

LOOK FOR

- The T wave ends well past halfway between consecutive R waves
- Broad, sometimes notched T waves
- Measure QT in II or V5 and correct for rate (QTc)

ACT

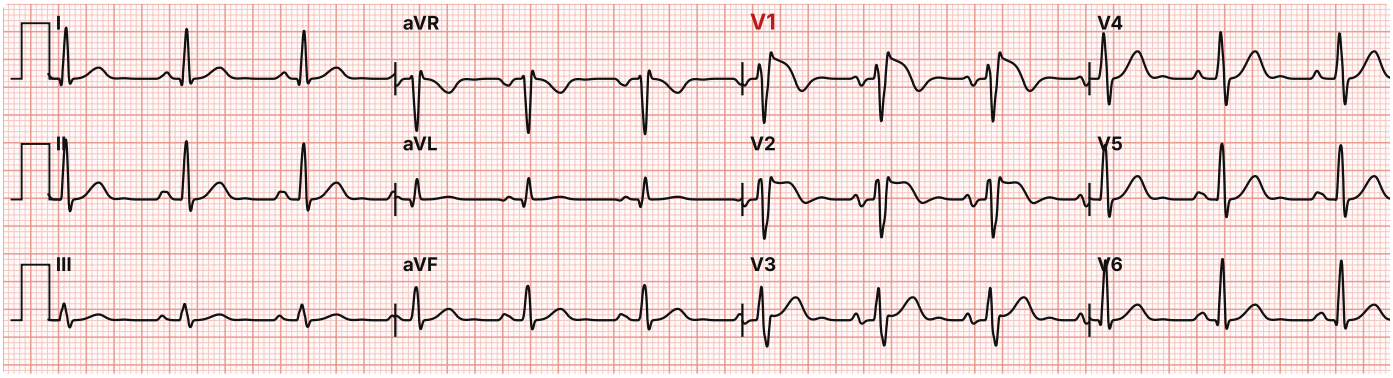
- **Stop QT-prolonging drugs** — Antiarrhythmics, antipsychotics, ondansetron, macrolides, fluoroquinolones, methadone
- **Correct K^+, Mg^{2+} and Ca^{2+}** — Aim for high-normal potassium and magnesium

Don't confuse with ≠ **Hypocalcaemia** long ST, normal T ≠ **Hypokalaemia** U wave merges with T (long QU)

Brugada pattern (type 1) URGENT

Look at V1

Coved ST elevation ≥ 2 mm in V1–V2 descending into a negative T wave.



CRITERIA

V1–V2	Coved STE ≥ 2 mm followed by a negative T
QRS	RBBB-like terminal r′
Reciprocal	None

LOOK FOR

- High J-point takeoff in V1–V2
- Coved (convex) ST descending into an inverted T
- Confined to the right precordial leads

ACT

- **Expert referral** — ICD for survivors of arrest or patients with syncope
- **Treat fever promptly** — Fever unmasks the pattern and can trigger arrhythmia

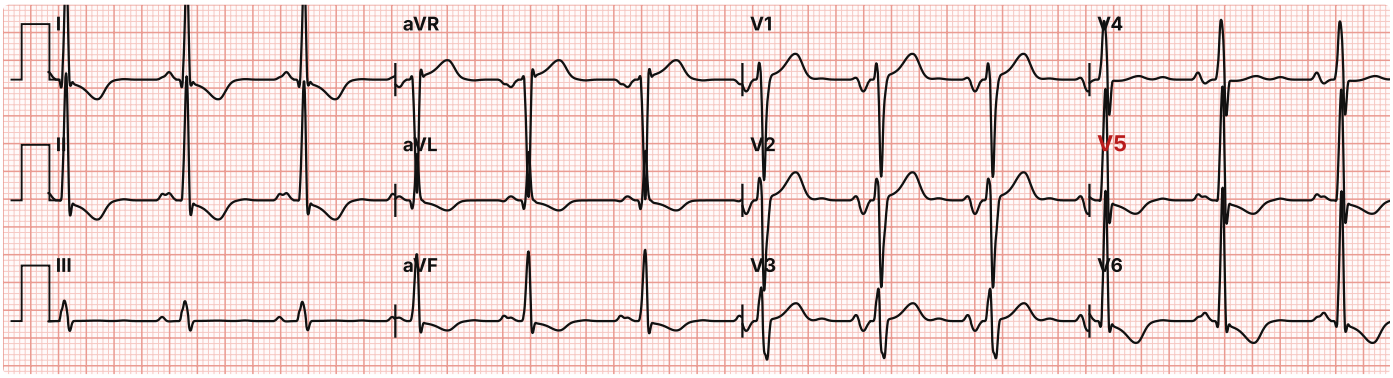
✕ **Avoid sodium-channel blockers**

Don't confuse with ≠ **Septal STEMI** evolves, with symptoms and often reciprocal change ≠ **RBBB** rSR′ without coved STE

Left ventricular hypertrophy with strain ASSESS

Look at V5

Very large LV voltages with lateral strain: down-sloping ST depression and asymmetric T inversion.



CRITERIA

Sokolow–Lyon	S(V1) + R(V5 or V6) > 35 mm
Cornell	R(aVL) + S(V3) > 28 mm (men), > 20 mm (women)
Strain	Down-sloping ST depression, asymmetric T inversion in I, aVL, V5–V6
P wave	Left atrial enlargement

LOOK FOR

- Deep S in V1–V2 and a tall R in V5–V6
- Down-sloping ST depression with asymmetric T inversion laterally
- Broad, notched P in II with a deep terminal component in V1

ACT

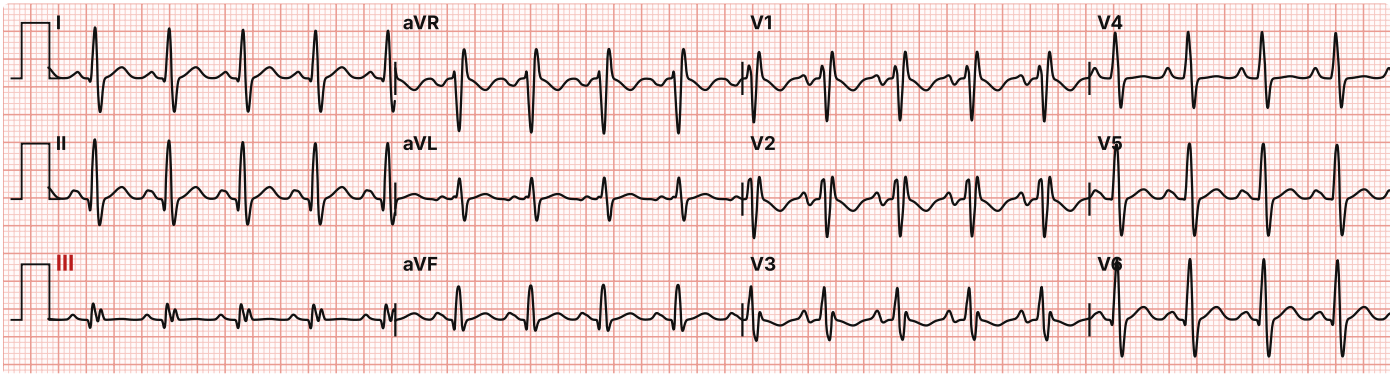
- **Clinical correlation and echocardiography** — Hypertension, aortic stenosis, cardiomyopathy

Don't confuse with ≠ **Ischaemia** depression without voltage criteria ≠ **Digoxin** scooped ST

Acute pulmonary embolism pattern URGENT

Look at III

Sinus tachycardia with right-heart strain: S1Q3T3, right precordial T inversion and RBBB.



CRITERIA

Rate	Sinus tachycardia (the most common finding)
S1Q3T3	S in I; Q and inverted T in III
RV strain	T inversion V1–V4
Conduction	Incomplete or complete RBBB
Axis	Rightward shift

LOOK FOR

- Sinus tachycardia
- S wave in I, Q wave and T inversion in III
- T inversion in V1–V4

ACT

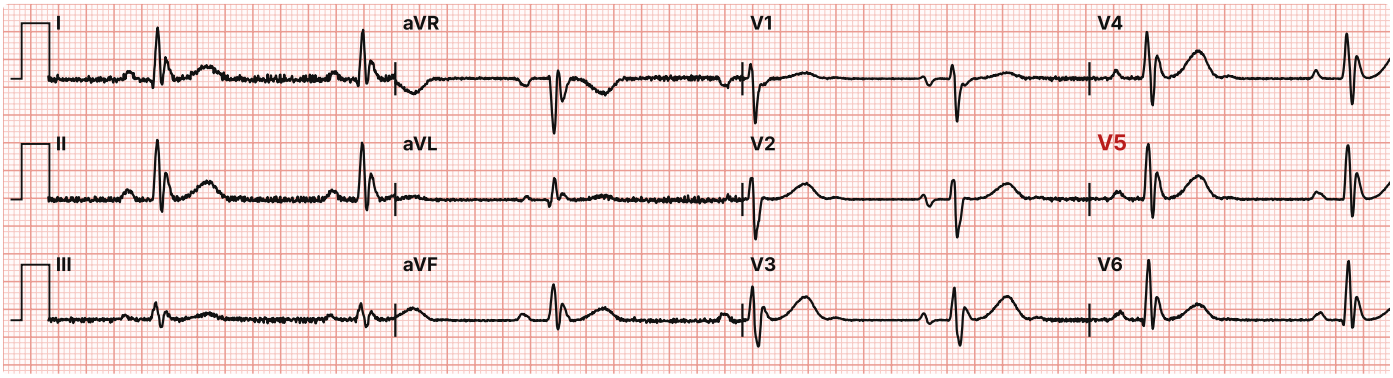
- **Massive PE with shock or arrest** — Thrombolysis or embolectomy
- **Oxygen, IV access, risk stratification** — Clinical probability scores
- **Anticoagulation** — As directed once PE is suspected and bleeding risk is acceptable

Don't confuse with ≠ **Sinus tach** Sinus tachycardia alone is the most common ECG in PE ≠ **Wellens** anterior T inversion from LAD stenosis

Hypothermia URGENT

Look at V5

Bradycardia, long intervals, Osborn (J) waves and shivering artefact.



CRITERIA

Rate	Sinus bradycardia (or AF / junctional)
Intervals	Prolonged PR, QRS and QT
Osborn wave	Positive hump at the J point, larger as temperature falls
Artefact	Shivering tremor

LOOK FOR

- A hump at the end of the QRS (Osborn wave), best in II and V3–V6
- Long PR and QT
- Fine tremor on the limb leads — the chest leads stay cleaner

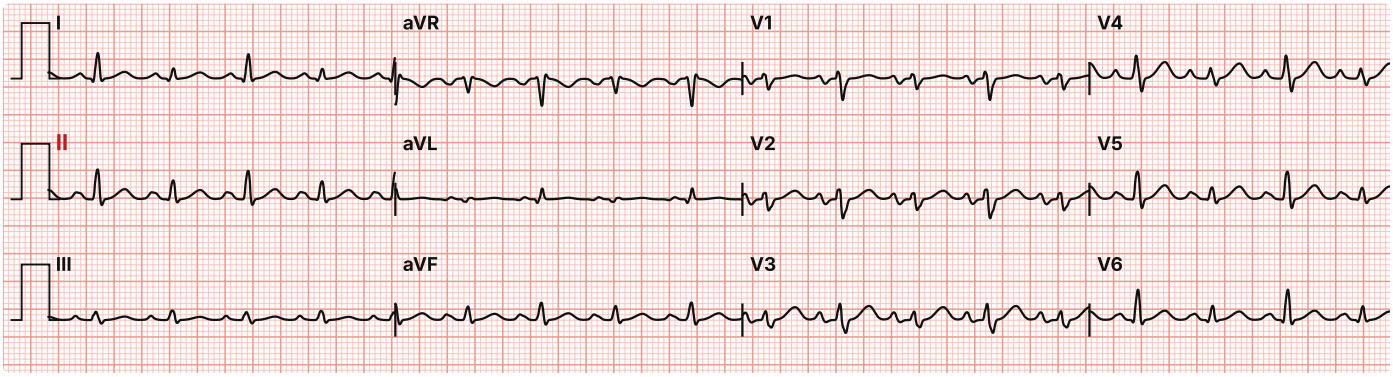
ACT

- **Rewarm** — Passive or active external rewarming

Don't confuse with ≠ **Sinus brady** Sinus bradycardia without J waves ≠ **Brugada** coved STE in V1–V2

Electrical alternans (tamponade) URGENT

Low-voltage sinus tachycardia whose QRS amplitude alternates beat to beat — the heart swinging in a large effusion.



CRITERIA

Rate	Sinus tachycardia
Voltage	Low (< 5 mm QRS in limb leads)
QRS	Amplitude and axis alternate beat to beat

LOOK FOR

- Every other QRS is smaller
- Low voltage throughout
- Tachycardia

ACT

- **Urgent echocardiography** — Confirm effusion and tamponade physiology
- **Pericardiocentesis for tamponade** — IV fluid as a temporising measure

Don't confuse with ≠ **Pericarditis** diffuse STE with PR depression ≠ **Sinus tachycardia** normal voltage

PRACTISE EVERY RHYTHM LIVE

The SmartCare ECG Simulator plays all 68 rhythms in this reference, with a 12-lead view, calipers and a quiz mode.

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solimananas.github.io/SmartCare ECG Simulator

A study aid, not a substitute for clinical judgement. Local protocols and medical direction always take precedence.